



ACQLERATE

MODULE 12

The Startup On-Ramp: SBIR, OTs, DIU, and the Valley of Death

How a startup, commercial company, or non-traditional supplier gets its first defense dollar and turns it into a business: the landscape of on-ramps, SBIR/STTR through Phase III, Other Transactions and consortia, DIU and the service innovation units, crossing the valley of death to a program of record, getting contract-ready, pricing without a cost history, and the capital structure rules that can end eligibility.

8 Lessons · ~184 Minutes · Lesson Book Edition

Course Lessons

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THE STARTUP ON-RAMP: SBIR, OTS, DIU, AND THE VALLEY OF DEATH · LESSON 1

The Non-Traditional Landscape: Why DoD Wants You and Why It Is Still Hard

20 min

Who counts as a non-traditional defense contractor, why the Department has spent a decade building on-ramps for startups and commercial firms, what those on-ramps actually are (SBIR/STTR, Other Transactions, DIU, the service innovation units, Commercial Solutions Openings), why the valley of death still swallows most of them, and what a first year in defense realistically looks like.

Two True Things at Once

The Department of Defense wants commercial technology and new suppliers more than at any time since the Second World War. It has said so in every strategy document for a decade, built organizations whose whole purpose is to find companies like yours, and created contracting tools that skip most of the paperwork. It is also true that the large majority of startups that take a first defense dollar never take a recurring one, and that the median time from first contact to a production contract, for the companies that make it, is measured in years. Both things are true because the front door and the back door are run by different people with different money. This module is about getting through both.

Who Counts as Non-Traditional

The law defines a non-traditional defense contractor as an entity that has not, for at least the past year, performed on a DoD contract or subcontract subject to full coverage under the Cost Accounting Standards. In practice that means most startups, most commercial companies, and many small businesses that have only done fixed-price or SBIR work. The status matters because it unlocks things: a prototype OT can be awarded without cost share if a non-traditional participates significantly, some cost and pricing data requirements are relaxed, and several programs exist specifically for you. Check the definition carefully before claiming it. A company that has done a cost-type subcontract for a large prime may already have lost the status without noticing.

Why the Department Built On-Ramps

The reasons are practical, not charitable. The technologies the Department most needs (software, autonomy, space, microelectronics, AI, biotech, energy) are led by commercial firms that will not rewrite their business to follow the FAR. The traditional acquisition system takes years to award and decades to field, and adversaries are not waiting. Congress responded with SBIR reauthorizations, expanded OT authority in the 2016 and later defense bills, and the creation of DIU, the service innovation units, and dedicated funds for transitioning prototypes. The Department responded by building organizations staffed with people

whose performance is measured by how many commercial companies they bring in. Those people are real, well-funded, and on your side. They are also not the people who will buy from you at scale.

The On-Ramps at a Glance

ON-RAMP	WHAT IT IS	TYPICAL FIRST AWARD	BEST FOR
SBIR/STTR	Phased R&D set-aside for small businesses	Phase I: low to mid six figures; Phase II: low seven figures	Small businesses with a technology to mature; the widest door
Prototype OT (direct or via consortium)	Non-FAR agreement to build and demonstrate a prototype	Varies widely, often seven figures	Companies with a near-ready product a program wants to try
DIU CSO	Pitch-based competition for commercial solutions to a stated problem	Prototype OT, often seven figures, 12 to 24 months	Commercial products adaptable to a defense use
Service innovation units (AFWERX, Army xTech, NavalX, SOFWERX)	Open topics, challenges, accelerators, matching funds	SBIR open topic Phase I or II; prize money	Startups looking for a first customer inside a service
Consortia (e.g., for Army, Navy, space, munitions, medical)	Membership groups through which OT prototype projects are competed	Project awards from five to eight figures	Companies that want a steady flow of prototype opportunities in one domain
Traditional FAR contract or subcontract	The regular system, including as a sub to a prime	Any size	Companies ready to comply; the eventual destination for most

The Valley of Death, Explained Simply

A Phase II SBIR or a prototype OT is paid from research and development money that an innovation organization controls. A production contract is paid from procurement or operations money that a program office controls, and that money is planned two years ahead through the budget process. When your prototype succeeds, the innovation organization has done its job and has no more money for you. The program office that could buy at scale did not plan for you two years ago, has no requirement written the way your product works, and has a budget already committed to other things. That gap, between the R&D dollar that found you and the procurement dollar that could sustain you, is the valley. Nothing about it is personal. It is the structure of how the money moves, and the entire second half of this module is about crossing it.

IN PLAIN TERMS

Picture a minor-league player spotted by a scout. The scout is genuinely impressed and writes a glowing report. But the scout does not sign paychecks. The team's general manager does, and this season's payroll was committed before anyone saw the player. Defense works the same way: innovation offices (the scouts) find you with research money, while program offices (the general managers) hold the procurement money and set their budgets about two years ahead.

REAL TALK

The innovation units want to help you transition. They will tell you so, and they mean it. Here is the part nobody says out loud at the pitch day: a DIU project manager cannot write a requirement into a PEO's budget, and an AFWERX contracting officer cannot make a wing commander buy your product with money the wing does not have. The people who found you are your advocates. They are not your buyers. Treat every prototype award as opening day of a two-year sales campaign to find whoever the buyer actually turns out to be.

The Shape of the Funnel**Thousands**

Phase I awards per year across DoD

The front door is wide

Under half

Of Phase I awardees reach Phase II

The first filter

A minority

Of Phase II efforts reach Phase III or production

The valley; estimates vary by service and year

2 to 4 years

Typical first-contact to production for those who cross

Plan cash accordingly

What Year One Realistically Looks Like

A company entering with a relevant product and no defense history can reasonably expect, in its first twelve to eighteen months, to register (SAM, DSIP, a CAGE code), win one or two Phase I SBIRs or a small prototype effort, meet a dozen government people who like the product, and learn which program office might eventually own the requirement. It can reasonably expect not to see a production order. Companies that plan for this (a commercial revenue base or investor runway that covers two to three years of defense pursuit, a founder or hire who spends real time inside the customer, and a disciplined view of which programs have money) tend to cross. Companies that treat a Phase I as a sales close, or that build the defense strategy around one enthusiastic officer, tend not to.

One sentence to teach your team: **the front door to defense is wide open and staffed by people paid to bring you in, but the money that sustains a company lives in program offices that planned their budgets two years ago, and the whole game is getting from the first to the second before the runway ends.**

INTERMEDIATE

Reading the Ecosystem: Who Has Money and Who Has Need

The ecosystem sorts into three kinds of organizations. Finders (DIU, AFWERX, xTech, SOFWERX, the SBIR program offices) have R&D money, fast contracting, and a mandate to bring companies in. Users (operational units, wings, brigades, combatant commands) have the need and the urgency but usually little procurement money of their own and no authority to start a program. Buyers (PEOs and program offices, and for sustainment the depots and logistics commands) have the recurring money and the requirement authority, and move on the budget calendar. The mistake most entrants make is spending all their time with finders and users because those meetings are easy and encouraging. The company that crosses the valley spends its second year with buyers, armed with evidence from the users and money from the finders.

Questions to Ask Every Government Contact in Year One

- **Who would buy this at scale?** Not who likes it. Which PEO or program office owns the mission this touches, and who there controls the budget line.
- **Is there a requirement?** A written, validated need that your product could satisfy. If not, who could write one, and how long would it take.
- **What appropriation would pay for it?** RDT&E for development, procurement for buying units, O&M for services and sustainment. Different money, different timing, different owners.
- **What is the transition path from this award?** A specific mechanism (Phase III, follow-on production OT, an existing vehicle), not a hope.
- **Who else is doing this?** Competing efforts inside the government, incumbents with the current solution, primes with a competing approach.
- **What would make you say no?** Security, integration, data rights, sustainment, or pricing objections surface earlier when asked directly.

Dual-Use as a Survival Strategy

The on-ramps are designed around companies that do not need the defense dollar to survive. That is not an accident: a company with commercial revenue can absorb DoD's timelines, keep its product roadmap on commercial terms, and negotiate data rights from strength. A company whose only customer is DoD ends up shaped by DoD, with product decisions driven by the next SBIR topic rather than a market. Investors know this, which is why serious defense-focused venture funds look for a commercial or at least a multi-agency story. If your product is defense-only by nature, the substitute for commercial revenue is a deliberately diversified government pipeline (several services, several programs) so that no single office's budget decision can end the company.

The Prime Route: Entering as a Subcontractor

The on-ramps get the attention, but the oldest path into defense is still open: become a subcontractor to a prime that already has the program. The prime has the contract, the customer relationship, the security infrastructure, and the compliance systems; you bring the technology. The economics are worse (the prime takes margin and often the customer relationship), the data rights negotiation is harder (primes want to own what goes into their system), and you will learn less about the government. But you can be earning on a program of record within months, not years, and primes have subcontracting plan obligations that make them want qualified small businesses. Many companies run both routes at once: direct on-ramp awards to build the government relationship, and a prime subcontract to earn on a fielded program.

ADVANCED

Why the On-Ramps Keep Getting Rebuilt

Every few years a new office, fund, or authority is created to fix the valley of death, and each one helps at the margin without closing it. The reason is structural: the valley exists because of the two-year budget cycle and the separation between R&D money and procurement money, and no innovation office can change either. The tools that actually move companies across are the ones that put money on the buyer's side of the valley: dedicated transition funds that let a program office buy before its own budget catches up, SBIR Phase III authority that lets a program office award without competition, and OT follow-on production authority that turns a successful prototype directly into a production agreement. Learn those three mechanisms deeply. The offices and acronyms will change; the money mechanics will not.

The Data Rights Position You Establish on Day One

Your first defense award sets a data rights precedent that follows the product. Under SBIR, the government gets a protected period of limited rights in the data you generate, and your pre-existing technology stays yours if you assert it correctly. Under an OT, data rights are negotiated freely, and a company that gives away rights to win a prototype has given away its position for the production negotiation. Under a FAR contract, the default rules depend on who paid for development, and the burden is on you to mark and assert. Senior founders treat the data rights clause of the first award as the most important term in it, because it decides whether the government can hand your design to a competitor for production, and whether a prime can absorb your technology into its system. The compliance module covers the mechanics; here the point is that the first award is where the precedent is set.

Four Company Archetypes and How Each Should Enter

The right on-ramp depends on what kind of company you are.

Venture-backed dual-use startup **SPEED**

Enter through DIU or a service open topic; use SBIR for non-dilutive runway; aim for a prototype OT with follow-on production language.

Investor runway covers the valley. The risk is losing focus chasing every SBIR topic; pick one or two mission areas.

Bootstrapped small technology firm **PATIENCE**

SBIR is the core engine; sequence Phase I to Phase II to Phase III deliberately; add a prime subcontract for cash flow.

Non-dilutive money is the advantage. The risk is becoming an SBIR mill that never transitions.

Established commercial company entering defense **LEVERAGE**

Prototype OT or CSO with the product as-is; resist customization; negotiate commercial terms and data rights from strength.

Commercial revenue funds the pursuit. The risk is underestimating security, integration, and sustainment requirements.

Services or integration small business **ACCESS**

The on-ramps are not built for you; enter through set-asides, vehicles, and prime subcontracts (see the small business module).

SBIR and OTs fund technology, not services. Do not spend a year chasing the wrong door.

WHAT SENIOR PMS DO DIFFERENTLY

Government innovation PMs who actually transition companies identify the buying program office before the prototype award, not after, and write the transition mechanism into the plan. Founders who cross the valley treat the finders as allies and the buyers as customers, spend year two in program offices, protect data rights from the first award, and keep enough non-defense runway that no single budget decision can kill the company.

BUILD ON THIS

- **OTAs & Streamlined Acquisition** The authorities behind the on-ramps
- **The PPBE Cycle** Why the buyer's money is two years away
- **Winning as a Small Business** The traditional machinery, for comparison

Key Terms

Non-Traditional Defense Contractor

A company that has not performed, for at least a year, on a DoD contract subject to full cost accounting standards coverage. The legal definition in 10 U.S.C. 3014 that unlocks OT eligibility and other flexibilities.

Valley of Death

The gap between a successful prototype or research contract and a funded program of record. Most non-traditional entrants die here, not at the front door.

SBIR/STTR

Small Business Innovation Research and Small Business Technology Transfer: congressionally mandated set-asides of federal R&D budgets awarded to small businesses in phases. The most common first defense dollar for a startup.

Other Transaction (OT)

An agreement under 10 U.S.C. 4021 or 4022 that is not a procurement contract, grant, or cooperative agreement, and so is not bound by the FAR. Used for research and prototypes, with a path to follow-on production.

DIU

The Defense Innovation Unit: the DoD organization built to field commercial technology fast, using Commercial Solutions Openings and prototype OTs, with offices near commercial tech hubs.

Service Innovation Units

AFWERX (Air Force), SpaceWERX (Space Force), Army Applied SBIR and xTech, NavalX and the Navy SBIR program, SOFWERX (SOCOM): the service-level organizations running open topics, challenges, and accelerators.

Commercial Solutions Opening (CSO)

A competitive procedure for buying innovative commercial solutions based on a short pitch and evaluation rather than a full FAR proposal. Used by DIU and increasingly across the services.

Program of Record

A funded acquisition program with a line in the budget, a program office, and a requirement. Where the recurring money is, and what every on-ramp is ultimately trying to reach.

Dual-Use

Technology with both commercial and defense customers. The business model most on-ramps are designed for, because it lets a company survive DoD timelines on commercial revenue.

Program Executive Office (PEO)

The organization that owns a portfolio of programs of record and their budgets. The customer with money, as distinct from the innovation unit that found you.

Knowledge Check

What makes a company a non-traditional defense contractor under the law?

- A. Being a startup
- ✓ **B. Not having performed, for at least a year, on a DoD contract subject to full Cost Accounting Standards coverage**
- C. Having fewer than 500 employees
- D. Never having a government customer

WHY

The definition is about CAS-covered contract history, not age or size. A cost-type subcontract can cost you the status.

What is the valley of death?

- A. The proposal writing phase
- ✓ **B. The gap between a successful prototype funded by R&D money and a production contract funded by a program office's procurement money**
- C. A security clearance backlog
- D. The protest period

WHY

The finder's money runs out when the prototype succeeds; the buyer's money was planned two years earlier.

Why did DoD build organizations like DIU and AFWERX?

- A. To replace the FAR
- ✓ **B. Because the technologies it needs are led by commercial firms that will not adopt traditional acquisition, and the traditional system is too slow**
- C. To reduce small business awards
- D. To manage sustainment

WHY

The on-ramps exist for practical reasons: access to commercial technology and speed.

Which on-ramp is the widest door for a small business with a technology to mature?

A. A GWAC

✓ B. SBIR/STTR

C. An 8(a) sole source

D. A CPFF contract

WHY

SBIR/STTR awards thousands of Phase I contracts a year and requires no prior defense history.

The innovation unit that awarded your prototype OT says it wants to help you transition. What can it not do?

A. Introduce you to program offices

✓ B. Write a requirement into a PEO's budget or make a program office buy with money it does not have

C. Award a follow-on prototype

D. Share your results

WHY

Finders are advocates, not buyers. The budget and the requirement belong to the program office.

What should a company realistically expect in its first 12 to 18 months in defense?

A. A production contract

✓ B. Registrations, one or two small awards, a dozen supportive contacts, and identification of the program office that might own the requirement

C. A prime contract

D. An IDIQ

WHY

Plan cash and expectations around a two- to three-year pursuit.

Why are the on-ramps designed around dual-use companies?

A. Commercial companies are cheaper

✓ **B. A company with commercial revenue can survive DoD timelines, keep its roadmap on commercial terms, and negotiate from strength**

C. Defense-only companies are prohibited

D. It is a legal requirement

WHY

Commercial revenue is the runway across the valley. A defense-only company should diversify across services and programs instead.

What is the advantage of entering as a subcontractor to a prime?

A. Better margins

✓ **B. Earning on a program of record within months, using the prime's customer relationship and compliance infrastructure**

C. Owning the customer relationship

D. Avoiding data rights issues

WHY

The prime route trades margin and customer access for speed. Many companies run it alongside direct on-ramp awards.

Which three mechanisms actually move companies across the valley because they put money on the buyer's side?

A. Pitch days, accelerators, and hackathons

✓ **B. Dedicated transition funds, SBIR Phase III authority, and OT follow-on production authority**

C. Grants, prizes, and CRADAs

D. GWACs, BPAs, and schedules

WHY

Offices and acronyms change; these money mechanics are what let a program office buy before its own budget catches up.

Match the organization type to what it has:**CORRECT MATCHES**

Finder (DIU, AFWERX, SBIR office)	R&D money, fast contracting, mandate to bring companies in
User (operational unit, wing, command)	Need and urgency, little procurement money
Buyer (PEO, program office)	Recurring money and requirement authority, on the budget calendar
Prime contractor	The existing contract, customer relationship, and compliance systems

WHY

Finders have R&D money and a mandate; users have need; buyers have recurring money and requirement authority.

Order a realistic path from first contact to production:**CORRECT ORDER**

1. Register in SAM and DSIP; get a CAGE code
2. Win a Phase I SBIR or small prototype effort
3. Identify the PEO or program office that owns the mission and budget
4. Secure a transition mechanism (Phase III, follow-on production OT, transition fund)
5. Production or sustainment award from the program office

WHY

Register, win a small R&D award, find the buying program office, secure a transition mechanism, then production.

Why is the data rights clause of your first defense award so important?

- A. It sets the invoice terms
- ✓ **B. It sets a precedent that follows the product and decides whether the government can hand your design to a competitor or a prime can absorb your technology**
- C. It determines your NAICS code
- D. It is required for SAM registration

WHY

Assert and protect rights from the first award; the production negotiation starts from wherever the first award left you.

THE STARTUP ON-RAMP: SBIR, OTS, DIU, AND THE VALLEY OF DEATH · LESSON 2

SBIR and STTR: Phase I to Phase III, and the Sole-Source Power Most Companies Never Use

24 min

How the SBIR/STTR program works from a company's side: eligibility, the three phases and what each pays, conventional topics versus open topics, Direct-to-Phase II, the data rights protection that makes SBIR unusually founder-friendly, and Phase III, the authority that lets any government office award you a sole-source contract of any size with no competition, which is the whole point of the program and the part most awardees never reach.

What SBIR Actually Is

Every federal agency with a large R&D budget must set aside a slice of it (a bit over three percent for SBIR, a smaller slice for STTR) for competitive awards to small businesses. That money cannot be spent any other way, which is why it exists in good budget years and bad. DoD runs the largest program, with each service and several agencies (Army, Navy, Air Force and Space Force, DARPA, MDA, DHA, SOCOM, and others) publishing topics on their own schedules. The program is structured in three phases. The first two are funded by the set-aside. The third is funded by whoever wants the product, and that is where the program stops being a research grant and becomes a procurement advantage.

Eligibility in Plain Terms

You must be a for-profit U.S. business with 500 or fewer employees including affiliates, more than half owned and controlled by U.S. citizens or permanent residents (or by other qualifying small businesses, or under the rules for venture-backed firms covered later in this module). The principal investigator must be primarily employed by you during the effort. Work must be performed in the United States, with narrow exceptions. For STTR, at least forty percent of the work must be done by the small business and at least thirty percent by the partner research institution, under a written agreement that allocates intellectual property. Foreign ownership, affiliation through investors, and where your engineers physically sit are the three eligibility questions that trip up otherwise qualified companies. Verify each before proposing, because they are checked at award and can be challenged later.

The Three Phases

PHASE	PURPOSE	TYPICAL DOD SIZE AND LENGTH	FUNDED BY	COMPETITION
I	Feasibility and team	Low to mid six figures; about six months	SBIR/STTR set-aside	Competitive against a topic
II	Prototype development	Low seven figures; 18 to 24 months; sequential and enhancement awards possible	SBIR/STTR set-aside, plus matching funds from a customer for enhancements	Competitive among Phase I awardees on that topic (or D2P2)
III	Commercialization: production, further development, services derived from the SBIR work	Any size; any length	Any non-SBIR money: procurement, O&M, RDT&E, a prime's contract	None required; sole source to the SBIR firm

Conventional Topics and Open Topics

A conventional topic describes a specific problem (a sensor with certain performance, a coating for a certain environment) written by a government technologist who wants that thing. You propose a solution to that problem. An open topic describes a mission area and invites you to propose what you have. Open topics, started by AFWERX and adopted widely, are how most commercial startups enter: you do not need to find a topic that matches your product, you bring the product and show a defense use. The trade is that open topics attract far more proposals, and the Phase I is often designed as a customer-discovery exercise: your job during it is to find the government user who will sign a memorandum saying they want the Phase II. That letter is frequently a condition of Phase II selection.

Direct-to-Phase II

If you have already done the feasibility work, with your own money, an investor's, or a commercial customer's, several DoD components will let you propose directly to Phase II. You document the Phase I-equivalent results in the proposal. D2P2 is the right path for a company with a working product that needs adaptation, not for a company with an idea. It also skips the customer-discovery period that a Phase I gives you, so companies using D2P2 should already know who their government user is.

REVIEWERS ARE SCREENING FOR SBIR MILLS

The SBIR program has a reputation problem: companies that live on Phase I and II awards for years without ever selling anything. The program calls them SBIR mills and has added rules to screen them out (benchmarks for Phase II transition rates, limits on serial awardees). The reason it matters to you is that reviewers now look hard at whether you are building a business or collecting grants. A credible commercialization plan and an identified customer beat a better technical volume.

Phase III: The Part That Matters

Phase III is not an award the SBIR office makes. It is an authority that any federal office can use. If work derives from, extends, or completes your SBIR-funded effort, any agency (not just the one that funded the SBIR) may award you a contract for it, sole source, in any amount, using any appropriation, without a justification under the usual competition rules, because Congress declared SBIR itself to be the competition. Phase III can be a production contract, a services contract, a subcontract from a prime, or a follow-on development effort. It can be awarded years later. It is the single most powerful sole-source authority available to a small business, and the majority of SBIR awardees never use it, because they do not know it exists, do not tell program offices about it, or cannot connect their work to a buyer with money. The next lesson on the valley of death is largely about how to use it.

IN PLAIN TERMS

Imagine a cook who wins a formal cook-off held by one hotel. The prize is not just the trophy. It is a standing rule that any hotel in the chain may hire her, without holding another cook-off, to make dishes built on the winning recipe. SBIR Phase III works the same way: because Congress treats the SBIR award as the competition, any federal office can buy work derived from your SBIR sole source, in any amount, with any money it has.

EXAMPLE (ILLUSTRATIVE)

A 12-person sensor company wins a Navy SBIR Phase I for \$150,000 (a hypothetical amount) and spends about six months proving feasibility. It then wins a competitive Phase II for \$1,200,000 and spends two years building a prototype. Together, the set-aside has funded $\$150,000 + \$1,200,000 = \$1,350,000$ of development. The Navy SBIR office has no more money for the company. But an Army program office needs a similar sensor. The founder brings the Army contracting officer the SBIR award history and shows how the Army need extends the Phase II work. The Army awards an \$8,000,000 production contract as SBIR Phase III, sole source, with its own procurement money. No new competition and no justification were required.

One sentence to teach your team: **Phase I proves the idea, Phase II builds the prototype, and Phase III is a standing legal authority for any government office to buy the result from you sole source with any money it has, so the job from day one is finding the office that will use it.**

INTERMEDIATE

SBIR Data Rights: Why the Program Is Founder-Friendly

Under SBIR, technical data and software generated in the effort receive a protection period (the policy directive sets it at twenty years from award; verify current) during which the government may use the data for government purposes but may not release it outside the government or use it to have someone else manufacture the item. After the period, the government gets broader rights. Data you developed before the award, at your own expense, remains yours under the normal rules if you identify and assert it. This is stronger than the default under many FAR contracts and far stronger than an unnegotiated OT. The mechanics that make it real: mark every deliverable with the SBIR data rights legend exactly as the clause requires, list your pre-existing technology in the assertions table, and never let a modification or a follow-on contract quietly change the rights. The compliance module covers the general rules; SBIR is the one place where the small business starts with the advantage.

The Phase I Proposal, What Evaluators Actually Weigh

- **Technical merit and feasibility** Is the approach sound and is the risk understood. For open topics, does the product plausibly work in the defense use case.
- **Team** Has this team built things like this. The principal investigator's primary employment with you is checked.
- **Commercialization potential** Who will buy it and why. Even at Phase I, a named potential government user helps.
- **Defense need** Does the government technologist or the open-topic sponsor see a mission use. For open topics, a letter from a user is often decisive.
- **Cost realism** The budget matches the work; labor rates are plausible; the indirect rates are explained even if provisional.
- **Compliance** Page limits, required forms, the fraud/waste certification, foreign ownership disclosures, and the required registrations. Non-compliant proposals are not evaluated.

Phase II Enhancements and Matching Funds

Several components offer enhancements to Phase II when a government customer or a private investor puts in matching money: the SBIR office adds funds in proportion to what the customer or investor commits. The Air Force calls its versions TACFI and STRATFI (tactical and strategic funding increases), with

matching from a program office and, for STRATFI, private capital, producing combined awards in the eight figures. Other services have their own. These matter for two reasons. The money is large enough to finish a product. And the requirement to bring a program office's money forces exactly the buyer conversation that crosses the valley: a program office that has committed matching funds to your Phase II has already started spending its own budget on you.

STTR: When to Use It

STTR requires a research institution partner and gives it a formal role and a share of the work. Use it when the technology genuinely originates in or depends on university research, when the professor and lab are part of the team, or when you need the institution's facilities. The intellectual property agreement between you and the institution is required and must be settled before award; universities have standard positions on ownership and licensing that can take months to negotiate, and a company that leaves this to the last week loses the award. STTR budgets are smaller than SBIR and the program is smaller, but competition is lighter and the topics are often closer to basic research.

ADVANCED

Making Phase III Work: The Mechanics

A Phase III award requires three things. A contracting officer who is willing to use the authority (many have never done it; bring them the policy directive, the statute, and examples). Documentation that the work derives from, extends, or completes SBIR-funded work (keep a clean lineage: which SBIR awards, what they produced, how the new work builds on it). And a funded requirement in the awarding office. The award is made under the normal contract types, with a Phase III designation, and the SBIR data rights carry forward and the protection period restarts for new data. Primes can award Phase III subcontracts, and the government can direct them to. Where companies fail: they cannot show the lineage because the SBIR work and the product diverged; they let the awarding office run a competition instead of using the authority, and lose it; or they wait for the SBIR office to initiate, when the SBIR office has no Phase III money and no role.

The Ownership Rules for Venture-Backed Companies

Companies majority-owned by multiple venture capital firms, hedge funds, or private equity firms can be SBIR-eligible at agencies that have opted in (DoD has), under specific conditions: no single such investor may own more than half, the investors themselves must be U.S.-controlled, and the company must disclose the structure. The eligibility questions are asked at each award, so a financing round can change your status between Phase I and Phase II. Foreign investor participation triggers additional disclosure under the 2022 reauthorization and can disqualify the company outright depending on the investor's country and the nature of the ties. Founders raising money should have the SBIR eligibility rules and the foreign investment rules in front of the term sheet, not after it.

Sequencing an SBIR Strategy Over Three Years

How companies that transition actually sequence the program.

Year 1: Two to three Phase I awards across two components ENTER

Open topics where the product fits; conventional topics only where the match is close. Use each Phase I to find a named government user and get a letter.

More than three Phase I efforts at once dilutes a small team. Fewer than two leaves the company hostage to one component's decision.

Year 2: One Phase II with matching funds BUILD

Pick the Phase I with the strongest user pull. Pursue an enhancement (TACFI-style) so a program office puts in money. Start the Phase III conversation with that office now.

Two Phase IIs in parallel is possible with a real team; it is a mistake for a five-person company.

Year 3: Phase III or follow-on production TRANSITION

Bring the program office and its CO the Phase III authority, the lineage, and a proposal. Alternatively, a prime subcontract designated Phase III.

If no office has money, the SBIR was research. Decide early whether to keep going or pivot the product.

WHAT SENIOR PMS DO DIFFERENTLY

Government SBIR program managers who transition technology pair each Phase II with a program office sponsor and matching funds from the start. Founders who succeed treat Phase I as customer discovery, use Phase II enhancements to get a buyer's money into the effort, keep the data rights assertions and the SBIR lineage clean, check eligibility before every financing round, and walk the Phase III authority into the buying office's contracting shop themselves rather than waiting to be found.

BUILD ON THIS

- **Data Rights and Intellectual Property** The general rules SBIR improves on
- **Affiliation** How investors affect size and SBIR eligibility
- **Crossing the Valley of Death** Using Phase III to reach a program of record

Key Terms

SBIR

Small Business Innovation Research: a set-aside of a fixed percentage of the extramural R&D budget of every large federal agency, awarded competitively to small businesses in phases. DoD is the largest participant.

STTR

Small Business Technology Transfer: the companion program requiring formal collaboration with a research institution (university, FFRDC, nonprofit research lab), with a smaller set-aside and its own work-share rules.

Phase I

Feasibility. A short effort (typically six months for DoD) at a modest fixed price to prove the concept and the team. Often the first defense dollar a company receives.

Phase II

Development. A larger award (typically up to two years) to build and demonstrate a prototype. Sequential Phase II awards and matching-fund enhancements can extend it.

Phase III

Commercialization. Work that derives from, extends, or completes SBIR-funded effort, funded by non-SBIR money. Any agency may award Phase III sole source, in any amount, without further competition, to the SBIR firm.

Open Topic

A solicitation that invites companies to propose any technology addressing a broad mission area rather than a narrowly specified problem. Pioneered by AFWERX; now used across the services.

Direct-to-Phase II (D2P2)

Authority allowing an agency to award Phase II to a company that can show Phase I-equivalent feasibility work done with its own or other funds, skipping Phase I.

SBIR Data Rights

The protection period (currently twenty years from award under the 2019 policy directive; verify current) during which the government may not disclose SBIR-generated technical data outside the government or use it for manufacturing by others.

DSIP

The Defense SBIR/STTR Innovation Portal: where DoD topics are published, questions are asked, and proposals are submitted.

Commercialization Plan

The Phase II proposal section describing how the technology will reach a market, government or commercial, and who will pay. Evaluated, and the difference between a research project and a business.

Knowledge Check

Where does SBIR money come from?

- A. A congressional earmark each year
- ✓ B. A mandatory set-aside of a fixed percentage of each large agency's extramural R&D budget
- C. Prime contractor subcontracting plans
- D. The Defense Production Act

WHY

The set-aside is mandatory, which is why the program exists in every budget year.

What is the purpose of Phase I?

- A. Production
- ✓ B. Proving feasibility and the team in a short, modest fixed-price effort
- C. Sustainment
- D. Certifying the accounting system

WHY

Phase I is about six months and is often a company's first defense dollar.

What makes Phase III different from Phases I and II?

- A. It is larger but otherwise the same
- ✓ B. It is funded by non-SBIR money and can be awarded sole source by any agency, in any amount, because SBIR itself counts as the competition
- C. It requires a university partner
- D. It is only for software

WHY

Phase III is a standing authority, not an SBIR office award. Most awardees never use it.

What is an open topic?

A. A topic with no page limit

✓ **B. A solicitation inviting any technology for a broad mission area rather than a narrowly specified problem**

C. A classified topic

D. A topic reserved for universities

WHY

Open topics let commercial companies bring what they have. Phase I under an open topic is often customer discovery.

When is Direct-to-Phase II the right path?

A. For any first-time applicant

✓ **B. When the company has already done feasibility work with its own or other funds and has a working product needing adaptation**

C. When the company has no product yet

D. When the topic is classified

WHY

D2P2 skips Phase I and its customer-discovery period, so the company should already know its government user.

Under SBIR data rights, during the protection period the government may:

A. Release the data to competitors

✓ **B. Use the data for government purposes but not disclose it outside the government or use it for manufacture by others**

C. Do nothing with the data

D. Sell the data

WHY

This is stronger than the default under many FAR contracts. It depends on marking deliverables correctly.

What does a Phase II enhancement like TACFI or STRATFI require?

A. A second Phase I

✓ **B. Matching money from a government customer (and for STRATFI, private capital), which the SBIR office then matches**

C. A university partner

D. A prime contractor

WHY

The matching requirement forces the buyer conversation that crosses the valley.

What is the STTR work-share rule?

A. 50/50

✓ **B. At least 40 percent by the small business and at least 30 percent by the research institution**

C. All work by the university

D. No requirement

WHY

STTR also requires a written IP agreement with the institution before award.

Why do most SBIR awardees never get a Phase III?

A. It is rarely legal

✓ **B. They do not know the authority exists, do not bring it to program offices, or cannot connect the work to a buyer with money**

C. The SBIR office refuses

D. Phase III requires a new competition

WHY

The SBIR office has no Phase III money and no role. The company must walk the authority into the buying office.

Match the phase to its funding source:**CORRECT MATCHES**

Phase I	SBIR/STTR set-aside
Phase II enhancement	Set-aside plus matching funds from a customer or investor
Phase III	Any non-SBIR appropriation or a prime's contract
STTR	A separate, smaller set-aside requiring a research institution partner

WHY

Phases I and II use the set-aside; enhancements add customer or investor matching; Phase III uses any non-SBIR money.

Order a three-year SBIR strategy:**CORRECT ORDER**

1. Win two or three Phase I awards across two components; find named users
2. Select the Phase I with the strongest user pull for Phase II
3. Secure a Phase II enhancement with program office matching funds
4. Bring the Phase III authority and lineage to the buying office's CO
5. Phase III production, services, or prime subcontract award

WHY

Enter with a few Phase I efforts, build one Phase II with matching funds, then transition through Phase III or production.

A venture-backed company raises a round between Phase I and Phase II. What should it check?

- A. Nothing; eligibility is set at Phase I
- ✓ **B. That no single VC owns more than half, investors are U.S.-controlled, and any foreign investor ties are disclosed and do not disqualify the company**
- C. That the round is under \$1M
- D. That the investors are small businesses

WHY

Eligibility is checked at each award. Financing can change it.

THE STARTUP ON-RAMP: SBIR, OTS, DIU, AND THE VALLEY OF DEATH · LESSON 3

Other Transactions and Consortia: Contracting Without the FAR

24 min

What an Other Transaction agreement is and is not, the prototype OT under 10 U.S.C. 4022 and its conditions (non-traditional participation, cost share, or a CO determination), the follow-on production authority that lets a successful prototype become a production agreement without competition, how consortia work and what membership buys, what an OT leaves unprotected, and how to negotiate one when there is no FAR clause to fall back on.

What an OT Is

The FAR is not a law of nature. It applies to procurement contracts, and Congress has given DoD authority to make agreements that are not procurement contracts: Other Transactions. An OT is a contract in the ordinary legal sense (a binding agreement, enforceable, with money changing hands), but because it is not a procurement contract under the statutes, the FAR does not apply, the competition rules do not apply in their usual form, cost accounting standards do not apply, and the standard clauses are absent unless the parties put them in. What replaces them is the agreement itself. Everything about the deal (payment, data rights, deliverables, termination, disputes, audit) is whatever the two sides negotiate and write down. That freedom is the point, and also the risk.

Research OTs and Prototype OTs

Research OTs (10 U.S.C. 4021) fund research and usually require the company to share cost; they are used by labs and DARPA for early-stage work. Prototype OTs (10 U.S.C. 4022) are what the on-ramps use, and the definition of prototype is deliberately broad: a pilot, a demonstration, a proof of concept, an agile development, a test of a commercial product in a military setting. A prototype OT can be awarded when at least one non-traditional defense contractor or nonprofit research institution participates significantly, or when all significant participants are small businesses or non-traditionals, or when at least one-third of the cost comes from non-federal sources, or when a senior official determines in writing that exceptional circumstances justify it. For a startup, the first condition is usually met by being the startup. Awards above set dollar thresholds require approval at progressively higher levels (verify current thresholds; they have risen over time).

Follow-On Production: The Reason to Care

The provision that turns an OT from a nice experiment into a business is follow-on production. If the prototype OT was awarded using competitive procedures, and the prototype is successfully completed, and the solicitation and the agreement both said that follow-on production could be awarded without further

competition, then the government may award a production contract or a production OT to you directly. No new competition, no J&A, no protest window for a competitor. This is how a two-million-dollar prototype becomes a hundred-million-dollar production agreement. It is also the term most often missing: companies sign prototype OTs that say nothing about follow-on production, or whose success criteria are so vague that no one can determine success, and then discover that the production award requires a full competition after all.

EXAMPLE (ILLUSTRATIVE)

Two startups each win a \$2,000,000 prototype other transaction (OT) through competitive project solicitations in the same consortium. Startup A read the solicitation and the agreement before signing. Both say follow-on production may be awarded without further competition, and success is defined as three measurable results in a field demonstration. The prototype meets all three, the government documents successful completion, and the program office awards a production agreement directly. Startup B signed the government's draft as written. The agreement is silent on follow-on production and its success criterion reads "demonstrates military utility." The prototype goes well, but nobody can agree it succeeded, and there is no production language. The production award has to be fully competed, and Startup B is now one bidder among several.

What to Check Before Signing a Prototype OT

- **Follow-on production language** Does the agreement (and the original solicitation) state that follow-on production may be awarded without competition. If not, ask why not and what the transition path is.
- **Success criteria** Measurable, agreed, and achievable within the period. Vague criteria mean a disputed determination later.
- **Data rights** There is no default. Whatever the agreement says is the deal. Silence favors no one and creates a fight.
- **Payment terms** Milestone payments, fixed amount or expenditure-based, invoicing mechanics, and what happens if a milestone slips.
- **Termination** Who can end it, on what notice, and what is paid on termination. There is no FAR termination clause to fall back on.
- **Disputes** No Contract Disputes Act coverage by default; the agreement should say how disputes are resolved.
- **Flow-downs and compliance** Which government requirements (security, export, cyber) the agreement imports. Fewer than a FAR contract, but not none.

Consortia: How Most Prototype OTs Are Actually Awarded

A government sponsor (an Army command, a Navy systems command, a joint office) signs one large OT with a consortium management firm, and companies join the consortium as members, usually for a modest

annual fee. The sponsor then issues project solicitations to members, members submit short white papers and then proposals, and awards are made as project agreements under the umbrella OT. There are dozens of consortia, organized by domain: ground vehicles, aviation, munitions, space, naval systems, medical, cyber, training, and more. For a company, membership buys a stream of solicitations in its domain, a simplified proposal process, and an award vehicle that already exists. The consortium takes a small assessment on awards. The catch is that the follow-on production question is answered at the project level, and the consortium's standard terms may or may not include it; read the base OT and the project agreement both.

AN OT IS A NEGOTIATION WITH NO REFEREE

An OT is a negotiation between two parties with no referee. A large prime with a legal department will negotiate an OT to its advantage. A startup that treats the government's draft as a form to sign will get the government's draft. The government AOs are not trying to take advantage of you, but they are not going to volunteer terms you did not ask for. Get someone who has negotiated OTs before to read yours, and treat data rights and follow-on production as the two terms worth spending money on.

Why OTs Grew

Tens of billions

Annual DoD OT obligations in recent years
From under a billion a decade ago

Dozens

Active consortia
Most prototype OT dollars flow through them

Months, not years

Typical time from solicitation to award
The speed the authority was built for

Negotiable

Every term
No FAR defaults; the agreement is the whole deal

One sentence to teach your team: **an OT is a contract without the FAR, which means every protection you would normally get from a clause has to be negotiated into the agreement, and the two terms that decide whether the prototype becomes a business are the data rights and the follow-on production language.**

INTERMEDIATE

Competitive Procedures and Why They Matter

Follow-on production without competition depends on the prototype having been awarded using competitive procedures to the maximum extent practicable. A consortium project solicitation open to all members counts. A CSO counts. A sole-source prototype OT, awarded because one company had the technology, generally does not, and a production award after it will need its own justification. For a company, this means the fast path to production runs through a competition you win, not a relationship that

skips one. If a sponsor wants to award you a prototype OT directly, ask how production will be handled afterward, and consider whether a competition you would win anyway is worth the delay for the production authority it unlocks.

OT vs. FAR Contract vs. SBIR, From the Company Side

FEATURE	PROTOTYPE OT	FAR CONTRACT	SBIR
Governing rules	The agreement and the statute	FAR, DFARS, the clauses	FAR contract with SBIR policy directive overlays
Data rights	Fully negotiable; no default	Default rules by funding source; assertions required	Statutory protection period; strong for the company
Cost accounting and audit	Only what the agreement requires	CAS and DCAA where thresholds apply	Usually fixed price, light audit
Path to production	Follow-on production if competed and stated	Normal competition or J&A	Phase III sole-source authority
Protest	Limited; not a procurement under GAO's usual jurisdiction, with narrow exceptions	Full protest rights	Full protest rights on the underlying contract
Speed	Fast	Slow	Medium; on the component's cycle

Data Rights in an OT: Starting From Zero

Because the DFARS data rights clauses do not apply unless imported, an OT starts with no rules. The government's draft will often propose something close to the DFARS default or broader. Your counter should start from the commercial position: the government gets a license to use the deliverables for the purpose of the project and for government evaluation; the company retains ownership and all commercial rights; anything developed entirely at your expense before or during the project remains yours; and government-purpose or unlimited rights attach only to specifically identified items the government pays to develop and only after negotiation. Agree on what is delivered (a working prototype, reports, test data) versus what is merely used (your source code, your design files), because rights attach to deliverables. The compliance module's data rights lesson explains the categories; in an OT, you use them as vocabulary in a negotiation rather than as rules that bind you.

What an OT Does Not Give You

The absence of the FAR cuts both ways. No termination for convenience clause means no defined recovery if the government walks away, unless you negotiate one. No Contract Disputes Act means no board of contract appeals route by default. No prompt payment rules unless imported. No standard changes clause, so scope changes are bilateral negotiations from scratch. No small business set-aside protections. No

CPARS by default, so a successful OT may not build the past performance record that FAR competitions ask for (ask for a performance assessment in the agreement). Companies that come from the commercial world find this normal; companies that come from FAR contracting are surprised by how exposed they are.

ADVANCED

Negotiating Success Criteria and the Transition Clause

The follow-on production authority turns on successful completion, and successful completion is whatever the agreement says. Senior negotiators write the criteria as a short list of measurable outcomes (performance thresholds, a demonstration in a stated environment, delivery of a stated quantity), each with the evidence that will prove it and who decides. They add a transition section: upon successful completion, the government intends to award follow-on production under 4022(f), the anticipated quantity or scope, the pricing basis (a negotiated price schedule, or a method for reaching one), and the timing. None of it binds the government to buy, but it binds the government to a process, and it gives the program office a document it can carry into its own budget planning. Without it, successful completion is a debate and follow-on production is a hope.

The Government Side: Why AOs Say No

Agreements officers turn down OT proposals for reasons companies rarely hear. The project is not really a prototype (it is services or production dressed as a pilot, which the statute does not cover). The non-traditional participation is not significant (a startup listed as a subcontractor for two percent of the work does not qualify the prime's project). The approval thresholds require sign-off the sponsor cannot get in time. The follow-on production request has no funded requirement behind it, so promising it would be empty. Or the sponsor is using the OT to avoid a competition it should have run. A company that understands these constraints proposes projects that fit the authority, documents its own non-traditional status, keeps the prototype scope honest, and helps the sponsor find the program office whose money makes the transition clause real.

Term by Term: A Startup's OT Negotiation Positions

Starting positions that experienced OT counsel bring to the table.

Data rights **PROTECT**

Company owns everything; government gets a project-purpose license to deliverables; broader rights only for specifically funded, identified items.

Separate deliverables from background technology. Rights attach to deliverables.

Follow-on production **TRANSITION**

Explicit 4022(f) language, measurable success criteria, anticipated scope and pricing basis, and a timeline.

Confirm the original solicitation also stated it; the statute requires both.

Payment **CASH**

Milestone payments front-loaded to cover ramp-up; payable on milestone completion, not on the government's acceptance timeline.

A startup cannot float the government for 90 days. Say so.

Termination and disputes **EXIT**

A convenience termination with recovery of costs incurred and a reasonable wind-down; a defined dispute path with escalation and a neutral option.

Absent these, a walk-away leaves you with nothing but a lawsuit.

Performance record **REPUTATION**

A written performance assessment at completion that can be cited as past performance in later competitions.

OTs do not generate CPARS by default. Ask.

WHAT SENIOR PMS DO DIFFERENTLY

Government sponsors who transition prototypes compete them, write measurable success criteria and a transition section into the agreement, and line up the program office and its money before the prototype ends. Founders who turn OTs into businesses negotiate data rights from the commercial position, insist on follow-on production language, front-load milestone payments, get a performance assessment in writing, and never sign the government's first draft.

BUILD ON THIS

- **OTAs & Streamlined Acquisition** The authorities in the acquisition system context
- **Data Rights and Intellectual Property** The vocabulary for the OT negotiation
- **DIU, AFWERX, and Commercial Solutions Openings** Where most prototype OTs start

Key Terms

Other Transaction (OT)

A legally binding agreement that is not a procurement contract, grant, or cooperative agreement, and so is not subject to the FAR, the Competition in Contracting Act, or most procurement statutes. Governed by its own terms and the authorizing statute.

10 U.S.C. 4021

Authority for research OTs: basic, applied, and advanced research projects, typically with cost sharing.

10 U.S.C. 4022

Authority for prototype OTs: projects directly relevant to enhancing mission effectiveness or improving platforms, systems, components, or materials. The authority most on-ramps use.

Prototype Project

Broadly defined: a preliminary pilot, test, evaluation, demonstration, or agile development of a technology, process, or system, including a proof of concept, a model, or a pilot of a commercial item in a defense use.

Non-Traditional Participation

The most common condition for a prototype OT: at least one non-traditional defense contractor or nonprofit research institution participates to a significant extent (or all significant participants are small businesses or non-traditionals).

Cost Share

The alternative condition: at least one-third of the total project cost is paid by parties other than the federal government.

Follow-On Production

Authority under 4022(f) to award a production contract or transaction, without further competition, for a prototype that was competitively awarded and successfully completed, provided the solicitation and agreement said so.

Consortium

A membership organization, run by a consortium management firm under an OT with a government sponsor, through which prototype projects are solicited and awarded to members. Dozens exist by domain.

Agreements Officer (AO)

The government official with authority to negotiate and sign OTs, the OT counterpart of a contracting officer. Often the same person, warranted separately.

Successful Completion

The determination, against criteria set in the OT, that the prototype met its objectives. The trigger for follow-on production and a term worth negotiating precisely.

Knowledge Check

What is an Other Transaction?

A. A grant

✓ **B. A binding agreement that is not a procurement contract, grant, or cooperative agreement, and so is not subject to the FAR**

C. A FAR contract with fewer clauses

D. An informal understanding

WHY

It is a real contract in the legal sense, governed by its own terms and the authorizing statute rather than the FAR.

Which condition most commonly qualifies a startup's project for a prototype OT?

A. Cost share of one-third

✓ **B. Significant participation by a non-traditional defense contractor, which the startup usually is**

C. A senior official's exceptional-circumstances determination

D. A university partner

WHY

Being a non-traditional participating significantly is the usual basis. Cost share and the determination are alternatives.

What does follow-on production authority allow?

A. A second prototype

✓ **B. A production contract or OT awarded without further competition after a competitively awarded prototype is successfully completed, if the solicitation and agreement said so**

C. Unlimited sole-source awards

D. Automatic Phase III

WHY

All three conditions matter: competitive award, successful completion, and prior notice in the solicitation and agreement.

Why are vague success criteria dangerous in a prototype OT?

A. They make the prototype harder to build

✓ **B. Follow-on production turns on successful completion, and vague criteria make that determination a dispute**

C. They increase cost share

D. They trigger DCAA audit

WHY

Write measurable criteria, the evidence, and who decides.

How does a consortium work?

A. It is a joint venture that bids as one company

✓ **B. A sponsor signs one umbrella OT with a management firm; members receive project solicitations and win project agreements under it**

C. It is a government office

D. It replaces SAM registration

WHY

Membership buys a stream of domain-specific solicitations and an existing award vehicle.

What is the data rights default in an OT?

A. DFARS 7013 applies automatically

✓ **B. There is none; the agreement is the whole deal, and silence creates a fight**

C. The government gets unlimited rights

D. The company keeps everything automatically

WHY

Negotiate from the commercial position and separate deliverables from background technology.

A sponsor offers a sole-source prototype OT. What should the company ask?

A. Nothing; accept immediately

✓ **B. How production will be handled, since follow-on production without competition generally requires the prototype to have been competitively awarded**

C. Whether DCAA will audit

D. Whether it counts as Phase III

WHY

A competition you would win anyway may be worth the delay for the production authority it unlocks.

Which protection does an OT not provide unless negotiated in?

A. Enforceability

✓ **B. A termination for convenience recovery, a dispute resolution path, and prompt payment**

C. Payment of any kind

D. Binding signatures

WHY

The absence of the FAR removes protections as well as burdens.

Why might a successful OT not help in later FAR competitions?

A. OTs are secret

✓ **B. OTs do not generate CPARS by default, so there may be no past performance record unless a performance assessment was negotiated**

C. FAR competitions ignore prototypes

D. OT work is not past performance

WHY

Ask for a written performance assessment in the agreement.

Match the term to the concept:**CORRECT MATCHES**

10 U.S.C. 4021	Research OT authority
10 U.S.C. 4022	Prototype OT authority
4022(f)	Follow-on production without competition
Agreements Officer	The official who negotiates and signs OTs

WHY

4021 is research; 4022 is prototype; 4022(f) is follow-on production; the AO signs OTs.

Order the path from consortium membership to production:**CORRECT ORDER**

1. Join the consortium in your domain
2. Respond to a project solicitation with a white paper, then a proposal
3. Negotiate and sign the project agreement with follow-on production language
4. Complete the prototype and obtain the successful completion determination
5. Follow-on production contract or OT awarded without further competition

WHY

Join, respond to a project solicitation, win the project agreement, complete successfully against stated criteria, then follow-on production.

Why do agreements officers reject OT proposals?

A. They prefer FAR contracts

✓ B. The project is not really a prototype, the non-traditional participation is not significant, approvals cannot be obtained, or the transition has no funded requirement behind it

C. OTs are limited to \$1M

D. The company is too small

WHY

Propose projects that fit the authority, document non-traditional status, and help the sponsor find the program office with money.

THE STARTUP ON-RAMP: SBIR, OTS, DIU, AND THE VALLEY OF DEATH · LESSON 4

DIU, AFWERX, and Commercial Solutions Openings: How the Innovation Units Actually Buy

22 min

The Defense Innovation Unit and the service innovation units as they work in practice: the Commercial Solutions Opening process from problem statement to pitch to prototype OT, what a winning pitch contains, how AFWERX and its siblings run open topics and pitch days, what these organizations can and cannot do for you after the award, and how to read an innovation unit's problem statement for the program office hiding behind it.

The Innovation Units, Briefly

DIU is the department-level unit, created to bring commercial technology into DoD at commercial speed. It does not have its own programs; it takes problems from DoD partners (a combatant command, a service, an agency), runs a competition to find commercial solutions, awards prototype OTs, and works to transition the results to the partner. The services built their own units with different flavors. AFWERX runs the largest SBIR open topic program and the matching-fund enhancements, with the Space Force's SpaceWERX alongside. The Army runs xTech prize competitions and an Applied SBIR program tied to its program offices. The Navy has NavalX tech bridges connecting companies to commands and a SBIR program that has emphasized transition. SOCOM has SOFWERX, which prototypes fast for special operations users. Each has a website, a mailing list, and a calendar of open solicitations, and the entry cost for all of them is a registration and a short submission.

The CSO Process, Step by Step

A CSO starts with a problem statement: a page or two describing what a DoD user needs to do and cannot, written to describe the outcome rather than a solution. Companies respond with a solution brief, typically five pages or a short slide deck, describing the product, its maturity, its commercial customers, and how it would address the problem. The government evaluates briefs on merit (not against each other in the traditional sense, but each against the problem) and invites the strongest to pitch: an oral presentation and questions, usually with the end user in the room. Selected companies are then asked for a full proposal, which is negotiated into a prototype OT. From problem statement to award, DIU aims for months, and often hits it. The whole process is designed so that a commercial company's existing materials, adapted, are enough to compete.

IN PLAIN TERMS

A commercial solutions opening (CSO) works much like a job search. The job posting describes the problem to be solved, not the exact person wanted. You send a short resume that shows what you have already done (the solution brief). If it stands out, you interview with the manager you would actually work for (the pitch, usually with the end user present). Then you negotiate the offer (the prototype agreement). Your existing materials, tailored, are enough to apply.

What a Winning Solution Brief Contains

- **The problem, restated in the user's terms** Show you understood the mission need, not just the technology keywords.
- **The product as it exists today** What it does, who uses it commercially, what it costs, how mature it is. Not what it could become.
- **The adaptation** Precisely what would change for the defense use: integration, environment, security, data. Be honest about the gap.
- **Evidence** Commercial customers, deployments, test results, certifications. Innovation units are allergic to slideware.
- **The prototype plan** What you would demonstrate, where, with which users, in how many months, for roughly how much.
- **The team and the company** Who does the work, that you are a non-traditional, that you can handle the security and compliance the prototype needs.

AFWERX Open Topics and Pitch Days

AFWERX's open topic SBIR solicitations run in cycles, invite any technology with an Air Force or Space Force use, and award Phase I contracts in volume. The Phase I is explicitly a customer discovery period: you are expected to find an Air Force user, get a memorandum of understanding from them, and show up to Phase II with a customer who wants the product. Phase II selection weighs that customer letter heavily. TACFI and STRATFI then match program office and private funds to scale the Phase II. AFWERX also runs pitch events and challenges where companies present in person, and has at times awarded contracts on the day. The other services' versions differ in detail but share the shape: a wide, cheap entry; a filter based on finding a real user; and larger money only when a program office puts skin in.

TWO KINDS OF PROBLEM STATEMENT

Innovation units publish problem statements for two reasons. Sometimes a program office with money brought them a real gap and is standing behind the solicitation. Sometimes the innovation unit thinks the problem is important and hopes a program office will care once a prototype exists. The first kind transitions. The second kind produces a nice demo. Before you spend a month on a solution brief, find out which kind you are looking at. Ask, in the question period or in the pitch: who is the transition partner, and what money have they committed?

EXAMPLE (ILLUSTRATIVE)

A 25-person company sells route-planning software to trucking fleets. It spots two DIU problem statements that fit its product. During the question period it asks each one: who is the transition partner, and what money have they committed? For Problem Statement 1, a named program office has committed funds to follow-on work. For Problem Statement 2, the answer is that partners will be sought once a prototype exists. The company writes one five-page solution brief, for Problem Statement 1, using its commercial customer list and deployment data as evidence. It pitches with an engineer and a live demo, with the end user in the room. A few months later it signs a prototype OT, and it begins meeting the program office during the prototype, not after.

DIU by the Numbers, Roughly

Months

Problem statement to prototype award

The process is built for speed

Hundreds

Prototype projects awarded to date

Across autonomy, space, cyber, AI, energy, human systems

A minority

Of projects reach production

DIU publishes transition rates; read them

Billions

In follow-on production value on transitioned projects

The prize when it works

What the Innovation Unit Does After the Award

A DIU project manager will help you navigate the prototype: introductions to users, help with security and access, support with the transition partner. AFWERX will connect you to the program office and help structure the enhancement. What none of them can do is create a requirement, put money in a program office's budget, or compel a command to buy. Their transition help is advocacy and connection. The transition itself is a sale you make to the program office, using the evidence the prototype produced and the authority (follow-on production, Phase III) the award created. Companies that understand this start the program office conversation during the prototype, not after it.

One sentence to teach your team: **the innovation units run fast, merit-based competitions that a commercial company can enter with adapted commercial materials, and the question that decides whether the award leads anywhere is which program office is standing behind the problem statement with money.**

INTERMEDIATE

Reading a Problem Statement for the Buyer Behind It

A problem statement is written to be solution-agnostic, but it is not customer-agnostic. Read it for the mission (whose job is described), the platform or environment (what it must integrate with, which points at a program office), the transition partner if named, and the language of the requirement (a problem statement that echoes a published capability need or a program's stated gap is one a program office helped write). Then look outside the solicitation: the same organization's budget justification documents, the program office's public statements, recent contract awards in the area, and who spoke at the industry day. The company that can name the PEO and the budget line before pitching is the one whose pitch talks about transition credibly.

The Innovation Units Compared

UNIT	INSTRUMENT	ENTRY	STRENGTH	WATCH FOR
DIU	CSO to prototype OT with follow-on production	Solution brief against a problem statement	Speed; department-level reach; explicit transition focus	Whether the partner has committed money
AFWERX / SpaceWERX	SBIR open topics; TACFI/STRATFI; challenges	Open topic Phase I proposal	Volume; matching funds that pull program offices in	Phase I without a user letter goes nowhere
Army xTech / Applied SBIR	Prize competitions feeding SBIR; program-tied topics	xTech submission or SBIR proposal	Direct ties to Army program offices	Prize money is small; the value is the follow-on
NavalX / Navy SBIR	Tech bridges; SBIR with transition emphasis	Engagement through a tech bridge; SBIR proposal	Command-level relationships	Naval integration and certification burdens
SOFWERX	Rapid prototyping events and OTs	Event participation; RFI responses	Fast, user-driven, small quantities	SOCOM buys small; scale lives in the services

The Pitch: What Evaluators Are Deciding

The pitch is not a sales presentation. The evaluators (a DIU project team, the end user, sometimes the transition partner) are deciding three things. Does the product actually work today, at the maturity you claimed, for customers who pay for it. Do you understand the defense use well enough to adapt it without being told everything. And can your company execute a government prototype: security, integration, the pace, the paperwork. Bring the product if you can (a live demonstration beats slides), bring an engineer who can answer technical questions, and bring a clear statement of what you need from the government (data, access, users, test environments) to make the prototype succeed. Companies lose pitches by overclaiming maturity, by treating the user's questions as objections, and by having no answer to the transition question.

Prize Competitions and Challenges

xTech, AFWERX challenges, and various department-level prize competitions award cash prizes (typically five to six figures) for solutions to stated problems, with no contract and no strings. The prize money is not the point. The point is that winners are usually invited to submit SBIR proposals with a strong evaluator preference, get exposure to program offices that judge the competition, and earn a credential that opens doors. Prize competitions are also the fastest way for a company with no government experience to learn how government evaluators think. Enter them for the access, budget the time honestly, and treat the prize as a marketing expense that pays for itself.

ADVANCED

Structuring the Prototype for Transition

Companies that transition from innovation unit awards design the prototype around the transition partner's decision. The demonstration environment is the partner's environment, or as close as security allows. The users in the prototype are from the partner's community. The metrics measured are the ones the partner's requirement or the program's key performance parameters use. The deliverables include what the partner needs for its own process: test data in its format, a security assessment that supports an authority to operate, a cost estimate the partner can put in a budget request, and a sustainment concept. The prototype OT's success criteria mirror the partner's acceptance criteria. When the prototype ends, the partner has a package it can act on rather than a demo it enjoyed.

When the Innovation Unit Is the Wrong Door

The innovation units are built for commercial products that need adaptation, and they are the wrong door for some companies. If your product is defense-specific with no commercial market, the units will still fund a prototype but the dual-use expectation will work against you at transition; a direct program office relationship or a consortium in your domain may serve better. If you sell services or integration, the units

have little for you. If your technology is early (a concept, a component), a lab or DARPA research OT or a conventional SBIR topic fits better than a CSO expecting a working product. And if the program office you need already has a vehicle and a requirement, the fastest path may be to compete on the vehicle or team with the prime, skipping the prototype entirely. The innovation units are one door among several; pick the one that leads to the room with money.

A DIU Project From the Inside

The phases of a typical DIU project and what the company should be doing in each.

Solicitation and brief WEEKS 0 TO 4

Identify the partner and its money; write the brief in the user's terms; ask the transition question in the Q&A.

Do not submit to problem statements with no identifiable partner unless the exposure alone is worth the effort.

Pitch and proposal WEEKS 4 TO 12

Demonstrate live; bring an engineer; state what you need from the government; propose measurable success criteria that mirror the partner's acceptance.

Negotiate data rights and follow-on production into the OT here, not later.

Prototype execution MONTHS 3 TO 18

Work in the partner's environment with the partner's users; produce the partner's transition package; start the budget conversation with the PEO.

The partner's budget request for production must be written during the prototype to land in time.

Successful completion and transition MONTH 18+

Obtain the written success determination; bring the follow-on production proposal to the partner's contracting office.

If the partner has no money yet, bridge with a Phase III, a transition fund, or an extended prototype while the budget catches up.

WHAT SENIOR PMS DO DIFFERENTLY

Innovation unit PMs who transition projects only publish problem statements with a committed partner, write the transition into the OT, and start the partner's budget request during the prototype. Founders who succeed pick solicitations by the money behind them, pitch with a working product and a transition answer, design the prototype around the partner's decision, and treat the innovation unit as the introduction to the customer rather than the customer.

BUILD ON THIS

- **Other Transactions and Consortia** The agreement a CSO produces
- **Market Research** How the government side finds you
- **Proposal Development & Win Strategy** The traditional proposal, for contrast

Key Terms

Commercial Solutions Opening (CSO)

A competitive procedure, authorized in statute and DoD guidance, for acquiring innovative commercial solutions through a merit-based evaluation of short submissions, without a full FAR solicitation. Usually results in a prototype OT.

Problem Statement

The short description of a mission need that a CSO or DIU solicitation publishes: what the user is trying to do, the current gap, and the desired outcome. Deliberately solution-agnostic.

Solution Brief

The first-stage CSO submission: a short document (often five pages or slides) describing the commercial solution, its maturity, and how it addresses the problem.

Pitch

The second stage: an oral presentation and question session with the government evaluation team and the end user, after which selected companies are invited to propose.

DIU

The Defense Innovation Unit, reporting to the Secretary of Defense, with offices in commercial tech hubs. Runs CSOs against problems brought by DoD partners and awards prototype OTs with follow-on production language.

AFWERX

The Air Force's innovation arm, running the largest SBIR open topic program, pitch events, and the TACFI/STRATFI matching-fund mechanisms. SpaceWERX is its Space Force counterpart.

Army xTech and Applied SBIR

The Army's prize competitions (xTech) that feed into SBIR awards, and its Applied SBIR program that ties topics to program office needs.

SOFWERX and NavalX

SOCOM's collaboration and rapid prototyping hub, and the Navy's network of tech bridges connecting commercial companies to naval commands.

Transition Partner

The program office, PEO, or command that DIU identifies as the eventual buyer for a project. Named in the best DIU solicitations; absent in the ones to be careful of.

Pitch Day

An event where companies present to government evaluators and, in the strongest versions, receive same-day award decisions and contracts.

Knowledge Check

What does DIU do?

- A. Runs its own weapons programs
- ✓ **B. Takes problems from DoD partners, competes commercial solutions through CSOs, awards prototype OTs, and works to transition results to the partner**
- C. Manages SBIR for all services
- D. Certifies commercial products

WHY

DIU has no programs of its own; the partner is the eventual buyer.

What is the first-stage submission in a CSO?

- A. A full FAR proposal
- ✓ **B. A short solution brief describing the product, its maturity, and how it addresses the problem**
- C. A cost proposal
- D. A SAM registration

WHY

Typically five pages or a short deck, evaluated on merit against the problem.

What is the Phase I of an AFWERX open topic really for?

- A. Building the product
- ✓ **B. Customer discovery: finding an Air Force user who will sign a memorandum wanting the Phase II**
- C. Security clearance processing
- D. Production

WHY

Phase II selection weighs the user letter heavily.

Before writing a solution brief, what should a company find out?

A. The page limit only

✓ B. Which program office is the transition partner and what money it has committed

C. The evaluators' names

D. The consortium fee

WHY

Problem statements backed by a program office with money transition; the others produce demos.

What can an innovation unit not do after awarding you a prototype?

A. Introduce you to users

✓ B. Create a requirement, put money in a program office's budget, or compel a command to buy

C. Help with security access

D. Support the transition partner relationship

WHY

Their transition help is advocacy. The sale to the program office is yours to make.

What are evaluators deciding during a pitch?

A. Whether the slides are polished

✓ B. Whether the product works today as claimed, whether you understand the defense use, and whether your company can execute a government prototype

C. Whether you are the cheapest

D. Whether you have a clearance

WHY

Demonstrate live, bring an engineer, and have an answer to the transition question.

Why enter a prize competition like xTech?

A. The prize money funds the company

✓ **B. For the access: SBIR preference, exposure to judging program offices, and a credential**

C. It is required for SBIR

D. To avoid SAM registration

WHY

Treat the prize as a marketing expense that pays for itself in access.

How should a prototype be designed for transition?

A. To maximize technical novelty

✓ **B. Around the transition partner's decision: their environment, users, metrics, and the deliverables their process needs**

C. To minimize cost

D. To satisfy the innovation unit only

WHY

The partner should end the prototype with a package it can act on.

Which company is the innovation units the wrong door for?

A. A dual-use software startup

✓ **B. A services or integration firm, or a company whose target program office already has a vehicle and a requirement**

C. A company with commercial customers

D. A non-traditional

WHY

The units fund commercial products needing adaptation. Other companies have better doors.

Match the unit to its signature mechanism:**CORRECT MATCHES**

DIU	CSO to prototype OT with follow-on production
AFWERX	SBIR open topics and TACFI/STRATFI matching
Army xTech	Prize competitions feeding SBIR awards
SOFWERX	Rapid prototyping for special operations users

WHY

DIU runs CSOs; AFWERX runs open topics with matching funds; Army runs xTech; SOFWERX prototypes fast for SOCOM.

Order the CSO process:**CORRECT ORDER**

1. Government publishes a problem statement
2. Company submits a solution brief
3. Selected companies pitch to evaluators and the end user
4. Invited companies submit a full proposal
5. Prototype OT negotiated and awarded

WHY

Problem statement, solution brief, pitch, full proposal, prototype OT.

When should the program office budget conversation start?

A. After successful completion

✓ B. During the prototype, because the partner's production budget request must be written then to land in time

C. At the pitch

D. Never; the innovation unit handles it

WHY

The two-year budget lag means waiting until the prototype ends adds two years.

THE STARTUP ON-RAMP: SBIR, OTS, DIU, AND THE VALLEY OF DEATH · LESSON 5

Crossing the Valley of Death: From Prototype to Program of Record

26 min

The transition problem as a money and requirements problem: why a successful prototype does not become a purchase, the two-year budget lag and how to work inside it, finding the program office that owns the mission and the budget line, the four transition mechanisms (Phase III, OT follow-on production, transition funds such as APFIT, and existing vehicles), the requirement question, and a twelve-month transition campaign plan.

The Problem Restated

Your prototype worked. The users loved it. The innovation unit wrote a glowing report. And nothing happened, because the money that paid for the prototype was research money controlled by people who have no more of it, and the money that could buy the product at scale is procurement or operations money controlled by a program office that built its budget two years ago without knowing you existed, against a requirement written without your product in mind. The valley of death is not a failure of enthusiasm. It is the gap between two budgets and two sets of decision-makers, and crossing it is a campaign with a plan, a calendar, and a target.

The Two-Year Lag, and How Money Actually Moves

The finance module explains PPBE in detail; here is the piece that matters for transition. Each year, a service builds a budget proposal (the POM) for the years starting about two years out. That proposal goes through the department, the President's budget, and Congress before it is money a program office can spend. So the budget a program office is executing today was shaped two years ago, and the first budget your product can be planned into, starting now, is two years away. In between, there are only three ways to get money: find it in an existing line that has flexibility (a program with a line for exactly your kind of thing, or with unobligated funds it can redirect), move it by reprogramming (possible within limits, politically costly, needs a senior sponsor), or get it from a bridge fund created for exactly this gap. Your transition plan needs a near-term path (existing money or a bridge fund) and a long-term path (the POM), and they run in parallel.

IN PLAIN TERMS

Imagine selling a new software tool to a large company. The team that tried it loves it, but their department's spending was locked in last year's budget. The earliest real purchase is a future budget, unless a manager finds spare money in an existing line or a special innovation fund steps in. Defense transition works the same way: you need a near-term money path now and a planned budget line later, pursued at the same time.

Finding the Buyer

The buyer is a program office or command with three things: the mission your product serves, a budget line that can pay for it, and a requirement (or the ability to write one) that your product satisfies. Start from the mission and work to the money. The users who tried your prototype belong to a command; that command is supported by program offices under a PEO; those program offices have budget lines listed in public justification books with dollar amounts by year. Read the justification books for the lines nearest your product. Read the program office's public statements, industry day briefings, and recent awards. Find the person who owns the line (a program manager, a product lead) and the person who writes the requirement (often in the requirements or capability development side of the command). Those two people, plus a contracting officer willing to use the transition authority, are the buyer.

The Four Transition Mechanisms

MECHANISM	WHAT IT DOES	WHO USES IT	WHAT YOU NEED
SBIR Phase III	Any office awards you sole source, any amount, any money, for work deriving from your SBIR	Any program office or prime, with a CO willing to use the authority	SBIR lineage; a funded requirement; a CO who will do it
OT follow-on production	Production contract or OT without competition after a competed, successful prototype	The prototype's sponsor or a program office	Competed prototype; success determination; the language in the solicitation and agreement
Transition funds (APFIT and service equivalents)	Procurement money from a central fund to buy and field before the program's budget catches up	Program offices that apply with your product	A sponsor program office; a fielding plan; eligibility as small or non-traditional
Existing vehicle	The program office buys through a contract it already has (IDIQ, prime's contract, schedule)	Any office with a suitable vehicle	A prime or vehicle holder willing to add you; scope that fits

REAL TALK

You need a champion, a government person willing to spend their own credibility on you. Go get one. Then remember that a colonel who loves your product and a program office that has budgeted for it are two completely different things, and startups die betting on the first one. Champions retire. Champions rotate. Champions get overruled by someone who has never heard of you. A budget line does none of those things. Use the champion to get the line, then treat the line as the customer.

The Requirement Question

Program offices buy against requirements. A requirement is a validated statement of need, and validating one is a process (each military service validates its own requirements since the joint JCIDS process was disestablished in August 2025, with command processes below that) that takes months to years. Your prototype most likely satisfies no existing requirement exactly, because it was built to a problem statement rather than a requirement. Three responses. Find an existing requirement your product satisfies well enough, and let the program office buy against it; this is the fastest path and the reason to read requirements documents early. Get the requirement written or amended, with the users who tried your prototype as the advocates; slow, but durable. Or use an authority that does not need a validated requirement in the traditional form (some rapid acquisition and software pathway processes are built for this) with a sponsor willing to run it. Ask the buyer early which of the three applies, because the answer sets the timeline.

One sentence to teach your team: **transition is a campaign to get a specific program office to spend a specific budget line on your product against a requirement it can point to, using a transition authority its contracting officer will sign, and it takes a near-term money path and a two-year POM path running at the same time.**

INTERMEDIATE

The Twelve-Month Transition Campaign

Companies that cross run something like this, starting the day the prototype is awarded. Months one to three: identify the buying program office and the budget line; identify the requirement situation; meet the program manager and the requirements owner; agree on what evidence the prototype must produce for their decision. Months four to six: produce that evidence in the prototype; help the program office draft its input for the next POM cycle (they need a cost estimate, a fielding concept, and a sustainment approach from you); identify the near-term money path (an existing line, a transition fund application, a reprogramming). Months seven to nine: the POM input is submitted; the transition fund application goes in with the program office as sponsor; the contracting authority is selected and the CO briefed. Months ten to twelve: successful completion determined; the near-term award (Phase III, follow-on production, order on a vehicle)

negotiated; the POM line is in the budget request for two years out. Slip any of these by a quarter and the whole timeline slips a year, because the budget cycle only comes around once.

What the Program Office Needs From You

- **A cost estimate they can defend** Unit cost, integration cost, sustainment cost by year, in the format their budget analysts use. Not a price list.
- **A fielding concept** How many, to whom, in what sequence, with what training and support. They cannot budget what they cannot describe.
- **A sustainment approach** Who maintains it, what spares and licenses cost, what the data rights allow. O&S is most of the lifecycle cost and the first question a PEO asks.
- **Security and integration evidence** Assessment results supporting an authority to operate; interface documentation; test data in their format.
- **A requirement crosswalk** Your product's capabilities mapped to the language of their requirement or capability need.
- **A contracting path** The authority (Phase III, follow-on production, vehicle) written up for their CO with the statute and examples.

Transition Funds: APFIT and the Service Programs

Congress created APFIT to put procurement money on the buyer's side of the valley: a program office sponsors a project to buy and field an innovative technology from a small or non-traditional company, and if selected, the fund pays for the initial procurement while the program office plans its own money for later years. Award sizes have been in the low eight figures per project, and the program has expanded since it began (verify the current year's criteria and funding). The services have created their own transition and rapid-fielding funds with similar logic. The company's role is to find and support a sponsoring program office; the application is theirs to make. The strongest applications pair a successful prototype with a program office that has already put a line in its POM for the out years, so the fund bridges rather than substitutes.

Buying Through a Prime

Sometimes the fastest transition is through the prime that already holds the program contract. The program office can direct or encourage the prime to add your product as a subcontract, funded from the existing contract's money, possibly designated as a Phase III subcontract to preserve your SBIR rights. You get to a fielded program fast; the prime gets an integration task and margin; the program office avoids a new procurement. The risks are the ones the prime economics lesson describes: the prime controls the customer relationship, may want your data rights, and may prefer to build a competing version once it understands your product. Mitigate with a clear teaming agreement, Phase III designation where available, and a direct relationship with the program office that the prime cannot mediate.

ADVANCED

Reprogramming and UFRs: The Political Paths

Within a fiscal year, a service can move money between lines up to thresholds without congressional approval and above them with notification or approval. A senior sponsor who wants your product fielded this year can find the money by reprogramming from a line that is underexecuting. It happens, and it is expensive in political capital, so it is reserved for things leadership genuinely wants. The unfunded requirements list is the other political path: each year the services send Congress lists of needs they could not fit in the budget, and Congress often funds some of them. Getting on a UFR list requires a command or service to name your capability as a priority. Neither path is something a company can do; both are things a company can make easy by giving the sponsor a fielding plan, a cost, and a story that leadership can repeat.

Why Program Offices Say No, and What to Do About Each

Program offices decline transitions for reasons that are rarely about the technology. No budget: the answer is the near-term path plus the POM. No requirement: the answer is the crosswalk to an existing requirement or a sponsor for a new one. Sustainment: they cannot see who maintains it or what it costs over ten years; the answer is a sustainment plan and data rights that let them compete support. Integration: it does not fit their architecture or their security posture; the answer is engineering during the prototype, not after. Incumbent: an existing solution and a prime with a contract; the answer is a coexistence plan or the prime route. Risk to their program: a new component threatens their schedule; the answer is a fielding plan that isolates the risk. Companies that anticipate these six and bring the answer to the first meeting are treated as partners; companies that discover them at the decision are treated as risks.

Running Two Money Paths at Once

The near-term and long-term paths, and how they interact.

Near-term: existing money THIS YEAR

A line with flexibility, an underexecuting program, a prime's contract with scope, or an O&M account for a service-like purchase.

Small initial quantities are fine. The goal is a fielded unit and a contract that exists.

Near-term: bridge fund NEXT YEAR

APFIT or a service transition fund, sponsored by the program office, timed to the fund's application cycle.

Learn the fund's calendar in month one. Missing the cycle costs a year.

Long-term: the POM **TWO YEARS OUT**

A line for your product in the program's budget request, built from your cost estimate and fielding plan, submitted in the program office's POM input.

This is the customer that survives rotation. Everything else is a bridge to it.

Political: reprogramming or UFR **SPONSOR-DRIVEN**

Only with a senior champion; a story leadership will repeat; a fielding plan ready to execute.

Do not plan on it. Be ready for it.

WHAT SENIOR PMS DO DIFFERENTLY

Government PMs who transition prototypes write the POM input during the prototype, apply for a bridge fund with the prototype's evidence, and pick the contracting authority before the success determination.

Founders who cross the valley find the budget line before the pitch, give the program office a cost estimate, fielding plan, and sustainment approach in their formats, bring the contracting authority to the CO themselves, run the near-term and POM paths in parallel, and never mistake a champion for a budget.

BUILD ON THIS

- **The PPBE Cycle** Where the two-year lag comes from
- **Sustainment and O&S Cost** The first question a PEO asks
- **Prime vs. Sub Economics** What buying through a prime costs you

Key Terms

Transition

The move from an R&D-funded prototype or pilot to a procurement- or O&M-funded purchase by a program office or command. The event the whole on-ramp system exists to produce and mostly does not.

Two-Year Lag

The fact that the budget a program office spends this year was built roughly two years ago in the PPBE process, so money for something new must be inserted into a future year's budget or found by moving existing money.

Program Element / Budget Line

The specific line in the budget justification books where a program's money sits, by appropriation and year. Finding your buyer means finding the line.

Requirement

A validated statement of need through a requirements process (since August 2025, each military service validates its own; command processes sit below that). Program offices buy against requirements; a prototype that satisfies no requirement has no buyer with a mandate.

APFIT

Accelerate the Procurement and Fielding of Innovative Technologies: a congressionally created fund that provides procurement money to buy and field innovative technologies from small and non-traditional companies, bridging the years before a program's own budget catches up. Verify current program status and criteria.

Reprogramming

Moving money between budget lines within the year, subject to thresholds and congressional notification. A way to fund a transition sooner than the budget cycle allows, at a political cost.

Unfunded Requirement (UFR)

A need a service or command identifies and submits to leadership or Congress as worthy of funding above the budget. A list your product wants to be on.

Program Objective Memorandum (POM)

The service's multi-year budget proposal built each year for the years starting two years out. The document a transition has to get into.

Champion

A government person with standing in the buying organization who wants the product enough to spend their own credibility on it. Necessary and not sufficient.

Existing Vehicle

A contract already in place (an IDIQ, a prime's program contract, a GSA schedule) through which a program office can buy your product quickly, often by ordering through the prime.

Knowledge Check

Why does a successful prototype often lead to no purchase?

- A. The technology usually fails
- ✓ B. The prototype was paid with R&D money the innovation unit no longer has, while the buying program office built its budget two years ago without the product and against a different requirement
- C. Users rarely like prototypes
- D. Prototypes are classified

WHY

The valley is a gap between two budgets and two sets of decision-makers.

What is the two-year lag?

- A. The time to get a clearance
- ✓ B. The budget a program office spends now was built about two years ago, so new money must be planned two years out or found by other means
- C. The length of a Phase II
- D. The protest period

WHY

This is why transition needs a near-term path and a POM path in parallel.

What three things define the buyer?

- A. A user, a colonel, and a demo
- ✓ B. A program office with the mission, a budget line that can pay, and a requirement the product satisfies or can be written to
- C. A prime, a consortium, and an OT
- D. DIU, AFWERX, and SOFWERX

WHY

Start from the mission and work to the money and the requirement.

What is APFIT?

- A. A security certification
- ✓ B. A congressionally created fund providing procurement money to buy and field innovative technologies from small and non-traditional companies before the program's own budget catches up
- C. A SBIR phase
- D. A type of OT

WHY

A program office sponsors the application; the fund bridges the years before the POM line arrives.

Why is a champion necessary but not sufficient?

- A. Champions are rare
- ✓ B. Champions rotate, retire, and get overruled; a budget line survives them
- C. Champions cannot sign contracts
- D. Champions only exist in the Army

WHY

Use the champion to get the line, then treat the line as the customer.

Your prototype satisfies no existing requirement exactly. What are the options?

- A. Give up
- ✓ B. Crosswalk to an existing requirement, get one written or amended with the users as advocates, or use an authority that does not need a traditional validated requirement
- C. Wait for the innovation unit
- D. Ask for a bigger prototype

WHY

Ask the buyer early which applies, because it sets the timeline.

What does a program office need from you to write a POM input?

A. A press kit

✓ **B. A defensible cost estimate, a fielding concept, and a sustainment approach in their formats**

C. A prototype video

D. A list of investors

WHY

They cannot budget what they cannot describe and defend.

What is the main risk of transitioning through a prime?

A. Slower fielding

✓ **B. The prime controls the customer relationship, may want your data rights, and may build a competing version**

C. Higher prices for the government

D. No Phase III designation possible

WHY

Mitigate with a teaming agreement, Phase III designation where available, and a direct program office relationship.

Why is missing a budget or fund cycle by a quarter so costly?

A. It is not

✓ **B. The cycle comes around once a year, so a quarter's slip becomes a year's delay**

C. It voids the OT

D. It triggers a recompetete

WHY

Learn the calendars in month one.

Match the objection to the answer:**CORRECT MATCHES**

No budget	Near-term money path plus a POM input
No requirement	Crosswalk or a sponsor for a new one
Sustainment unclear	A ten-year sustainment plan and data rights allowing competed support
Integration risk	Engineering done during the prototype, not after

WHY

Anticipate the six standard objections and bring the answer to the first meeting.

Order the twelve-month transition campaign:**CORRECT ORDER**

- Months 1 to 3: identify the program office, budget line, requirement situation; agree on decision evidence
- Months 4 to 6: produce the evidence; help draft the POM input; pick the near-term money path
- Months 7 to 9: POM input submitted; transition fund application filed; CO briefed on the authority
- Months 10 to 12: successful completion; near-term award negotiated; POM line in the request

WHY

Identify buyer and requirement; produce evidence and draft POM input; submit POM and fund applications; complete and award.

What is reprogramming?

A. Rewriting the software

✓ **B. Moving money between budget lines within the year, subject to thresholds and congressional notification, at a political cost**

C. A new SBIR topic

D. Changing the contract type

WHY

A sponsor-driven path. Be ready for it; do not plan on it.

THE STARTUP ON-RAMP: SBIR, OTS, DIU, AND THE VALLEY OF DEATH · LESSON 6

Getting Contract-Ready: Registrations, Systems, and the Compliance Floor for a First Award

22 min

The unglamorous work that decides whether a first award can actually be made to you: SAM.gov and the UEI, the CAGE code, NAICS and size, DSIP and the SBIR company registry, banking and invoicing setup, what an adequate accounting system means and when you need one, the minimum cybersecurity posture for handling government information, foreign ownership and FOCI questions, and the order to do it all in so nothing blocks the award.

Why This Lesson Exists

More first awards are delayed by registration and system problems than by anything in the proposal. A SAM registration that lapsed, a CAGE code tied to a former address, a size representation that conflicts with the investor structure, an accounting system that cannot support the contract type the government wants to use, a cybersecurity score that was never posted: each of these has held up awards for months and killed some outright. None of it is hard. All of it takes longer than founders expect, and some of it has dependencies that must be done in order. Do this work before the first proposal, not after the first selection.

IN PLAIN TERMS

Think about starting a new job. The offer is signed, but you cannot be paid until HR has your tax forms, your identity documents and your direct deposit details, and each of those has its own process. Get your bank account number wrong and the first paycheck bounces. Government registrations are the same: until your entity is registered, validated, correctly represented and set up to invoice, a contracting officer cannot sign an award or send a payment.

SAM.gov: The Foundation

Registering in SAM means proving your entity exists (legal name, address, tax identification, and a validation of the entity that can take weeks if records do not match), receiving a UEI, being assigned a CAGE code by DLA, and completing the representations and certifications: your size under each NAICS code, ownership type, socioeconomic status, whether you are debarred or have certain compliance issues, and dozens of others. The registration is free (ignore the companies that charge to do it) and must be renewed every year; a lapse blocks awards and payments. The reps and certs are legal statements incorporated into every contract you sign. Read them, understand the ones about size and ownership in

particular, and update them when something changes, because a financing round or an acquisition can make yesterday's certification false.

The Registration Sequence

- **Entity basics** Legal name, physical address, EIN, and bank account exactly as they appear on official records. Mismatches stall validation.
- **SAM registration and validation** Submit; wait for entity validation and CAGE assignment; complete reps and certs. Allow four to eight weeks the first time.
- **SBA registrations** SBA company registry for SBIR/STTR; certifications for set-aside programs if eligible (see the small business module).
- **DSIP registration** Required to propose on DoD SBIR/STTR topics; separate accounts for the firm and for individuals.
- **PIEE / WAWF** Register to submit invoices; designate roles; test before the first invoice is due.
- **SPRS** Post the NIST SP 800-171 self-assessment score if any work will involve controlled information (most does eventually).
- **Innovation unit and consortium portals** DIU, AFWERX, and each consortium have their own sign-ups; do the ones in your domain.

EXAMPLE (ILLUSTRATIVE)

A 6-person startup plans to submit its first SBIR Phase I proposal in 10 weeks. Proposing requires a DSIP account, and DSIP needs an active SAM registration. Week 1: it starts SAM using its legal name, address and EIN. The address on file with the IRS does not match its new office, so entity validation stalls. Week 4: records are corrected. Week 7: validation completes, the CAGE code is assigned and the reps and certs are finished. That fits the four to eight weeks a first registration often takes, but leaves almost no margin. Week 8: it registers the firm and its people in DSIP. Week 10: the proposal goes in. If the company had waited until week 5 to start, it would have missed the deadline.

NAICS, Size, and Getting the Representation Right

Every solicitation carries a NAICS code with a size standard (employees or annual receipts). You represent in SAM, for each NAICS you work under, whether you are small. For a startup the number of employees or the receipts are usually not the issue; affiliation is. Investors with control, a parent company, common ownership with other firms, or heavy economic dependence on one customer can make you large by affiliation, and the small business module's affiliation lesson is required reading for any funded company. Get the representation right before the first set-aside or SBIR award, because a size protest or an SBA determination after award is expensive and public.

Banking, Invoicing, and Getting Paid

The government pays by electronic funds transfer to the account in SAM, after an invoice is submitted through WAWF and accepted by the government official designated in the contract. A first invoice typically takes longer than expected because roles were not set up, the receiving report was routed to the wrong person, or the invoice format did not match the contract's CLIN structure. Set up WAWF, identify who on the government side accepts, and submit a small first invoice early to find the problems. Understand the payment terms of the specific instrument: fixed-price SBIRs often pay on milestones or deliverables; OTs pay whatever the agreement says; cost-type contracts pay on cost vouchers with DCAA involvement. The business module's lesson on cash explains why this matters more for a small company than any margin question.

NOBODY WILL CHASE YOU TO FIX A REGISTRATION

Government contracting offices will not chase you to fix a registration. If SAM lapses the week before award, the award waits. If the SPRS score is missing, the CO cannot sign. If the bank account changed and SAM was not updated, the payment goes nowhere and you find out when it bounces back thirty days later. Assign one person to own the registrations, put every renewal on a calendar with a sixty-day warning, and check SAM status before every submission.

One sentence to teach your team: **the government can only award to an entity that is registered, validated, correctly represented, and able to invoice and protect information, and every one of those has a lead time, so build the compliance floor before the first proposal rather than after the first selection.**

INTERMEDIATE

Accounting Systems: When You Need One and What It Means

A fixed-price award (most Phase I SBIRs, many OTs) does not require a government-reviewed accounting system, though you still need to know your costs to price. A cost-reimbursement award (many Phase IIs, cost-type OT milestones, and most development contracts) requires an accounting system the government considers adequate before award: it must segregate direct costs from indirect costs, accumulate costs by contract and by task, allocate indirect costs through defined pools and bases, track labor by timekeeping that meets government expectations, exclude unallowable costs, and support interim billing. DCAA or DCMA reviews it, often using a standard form (verify the current version). Commercial accounting software can be configured to meet this, and several products are built for it; the harder part is the process discipline (timesheets daily, indirect pools defined, unallowables coded). Start before the Phase II proposal, because a proposal that says the system is adequate when it is not delays award and can be a false statement.

Which Requirements Apply to Which First Award

REQUIREMENT	FIXED-PRICE SBIR PHASE I	COST-TYPE PHASE II	PROTOTYPE OT	FAR SUBCONTRACT TO A PRIME
SAM, UEI, CAGE	Yes	Yes	Yes	Usually required by the prime
Adequate accounting system	No	Yes, before award	Only if the OT is cost-based and says so	If cost-type
SPRS / NIST 800-171 score	If CUI is involved	Almost always	If the agreement imports the clause; often does	Flowed down by the prime
WAWF invoicing	Yes	Yes	Depends; often yes	Prime's system instead
Reps and certs accuracy	Yes	Yes	Yes, as the agreement requires	Prime requires certifications
Facility clearance	Rarely	If classified	If classified	If classified; sponsored by the prime

The Cybersecurity Floor

If a contract involves federal contract information, the basic safeguarding requirements apply (a short list of practices). If it involves controlled unclassified information, which most technical work eventually does, the contract requires implementation of NIST SP 800-171, a self-assessment score posted in SPRS, and, as the CMMC program phases in, certification at the required level (verify the current phase-in status). A startup on commercial cloud tools can meet the basic requirements with configuration and policy; meeting 800-171 fully is a real project, and the score you post must be honest because it is a certification. Plan the cybersecurity work at the same time as the accounting system, because both take months and both block awards. The compliance module covers the details.

Foreign Ownership and the Questions That Come Early

Foreign ownership, control, or influence is formally assessed when a company needs a facility clearance for classified work, but the questions now arrive earlier: SBIR requires disclosure of foreign ties and investment and can disqualify on them; some OTs and CSOs ask; DIU asks; and the reps and certs include ownership questions. A startup with a foreign investor, a foreign-born founder with certain ties, offshore engineering, or a parent abroad should map these facts before proposing and understand which programs they affect. Some ties are disqualifying, some require mitigation, many are fine when disclosed. Concealing any of them is the one certain way to lose.

ADVANCED

Corporate Changes Mid-Award: Financing, Restructuring, and Novation

Startups change shape: a new round, a holding company, a merger, a conversion from LLC to corporation. Each can affect an active award. A financing can change size and SBIR eligibility. A change in the legal entity that holds the contract requires the government to recognize the successor through a novation agreement, which takes months and can be refused. A change of ownership triggers reps and certs updates and, on cleared contracts, FOCI review. Senior founders brief counsel on active government awards before any corporate change, keep the contracting entity stable where possible, and notify the contracting officer early rather than letting them discover it from a SAM update.

Building the Compliance Floor Cheaply

A five-person company cannot afford a compliance department, and does not need one for a first award. What it needs: one owner for registrations with a calendar; an accounting configuration and a written timekeeping policy (a day of a consultant's time if you use a product built for government contracting); a cybersecurity baseline on commercial cloud with policies written down and an honest self-assessment; a one-page map of ownership and foreign ties; and a folder of the certifications and assessments that every proposal asks for. The compliance module's lesson on building a program on a small team describes the full version; the on-ramp version is that folder, that calendar, and one person who reads the reps and certs.

The Award-Blocker Checklist

The things that stop a selected company from receiving its award, and when to fix each.

SAM lapsed or entity not validated FATAL

No award and no payment until fixed. Renew sixty days early; validate before the first proposal.

The most common and the most avoidable.

Size or ownership representation wrong SERIOUS

Award delayed or lost; potential false certification. Review after every financing.

Affiliation is the usual cause for funded companies.

Accounting system not adequate for cost-type MONTHS

Award converted to fixed price, delayed, or withdrawn. Start configuration before the Phase II proposal.

A proposal must not claim adequacy it does not have.

SPRS score missing BLOCKS SIGNATURE

CO cannot award a CUI contract. Post an honest score early.

Honest and low beats absent or inflated.

WAWF not set up BLOCKS PAYMENT

Award happens; cash does not. Register and test with a small first invoice.

Find the acceptor before the invoice is due.

WHAT SENIOR PMS DO DIFFERENTLY

Government contracting staff check SAM, SPRS, and the accounting system status before selection, not after, and tell companies early what will block. Founders who avoid the delays own the registrations with a calendar, verify size and ownership after every financing, start the accounting and cybersecurity work before the proposal that needs them, map foreign ties honestly, and brief counsel on government awards before any corporate change.

BUILD ON THIS

- **Affiliation** Why investors can make you large
- **CMMC and DFARS 7012** The cybersecurity requirements in full
- **Building a Compliance Program on a Small Team** The full version of the compliance floor

Key Terms

SAM.gov

The System for Award Management: the single federal registration every contractor must hold and renew annually to receive an award. Includes representations and certifications.

UEI

Unique Entity Identifier: the twelve-character identifier SAM assigns to your entity, replacing the old DUNS number. Required on every proposal and award.

CAGE Code

Commercial and Government Entity code: a five-character identifier assigned by DLA during SAM registration, used across DoD systems for your entity and location.

Reps and Certs

The representations and certifications you make in SAM (size, ownership, compliance status, exclusions) that are incorporated by reference into every award. False ones are a False Claims Act problem.

DSIP

The Defense SBIR/STTR Innovation Portal, where DoD SBIR topics are published and proposals submitted. Requires its own registration on top of SAM, plus the SBA company registry.

Adequate Accounting System

An accounting system that segregates direct and indirect costs, accumulates costs by contract, and supports billing, as required before a cost-reimbursement award. Determined by DCAA or DCMA review.

WAWF / PIEE

Wide Area Workflow within the Procurement Integrated Enterprise Environment: the DoD system through which invoices and receiving reports are submitted and approved. No WAWF, no payment.

FOCI

Foreign Ownership, Control, or Influence: the security review of foreign ties that applies when a company needs a facility clearance, and increasingly is asked about earlier through disclosure requirements.

SPRS Score

The self-assessed cybersecurity score under NIST SP 800-171, posted to the Supplier Performance Risk System, required before award of contracts involving controlled unclassified information.

Novation

The government's recognition of a successor entity on a contract after a merger, acquisition, or restructuring. Relevant when a startup's corporate structure changes mid-award.

Knowledge Check

What most often delays a first award?

A. Weak technical volumes

✓ **B. Registration and system problems: lapsed SAM, wrong representations, inadequate accounting system, missing cybersecurity score**

C. Protests

D. Price

WHY

None of it is hard; all of it has lead time and dependencies.

What are the reps and certs in SAM?

A. Marketing statements

✓ **B. Legal representations about size, ownership, and compliance status incorporated into every award; false ones are a False Claims Act problem**

C. Optional surveys

D. Bank details

WHY

Read them, and update them when a financing or acquisition changes the facts.

How much does SAM registration cost?

A. A few hundred dollars

✓ **B. Nothing; it is free, and companies that charge to do it are unnecessary**

C. It varies by state

D. A percentage of awards

WHY

Free, annual renewal, four to eight weeks the first time.

For a funded startup, what usually determines whether it is small?

A. Revenue alone

✓ **B. Affiliation through investors with control, parents, or common ownership, not just its own headcount**

C. The founder's preference

D. The NAICS code alone

WHY

Get the representation right before the first set-aside or SBIR award.

When is an adequate accounting system required?

A. For every award

✓ **B. Before a cost-reimbursement award; not for fixed-price**

C. Only for primes

D. Only after \$10M

WHY

Most Phase I SBIRs are fixed price; many Phase IIs and development contracts are cost-type.

What does an adequate accounting system do?

A. Files taxes

✓ **B. Segregates direct and indirect costs, accumulates by contract, allocates indirect costs through pools, tracks labor by compliant timekeeping, excludes unallowables, and supports billing**

C. Tracks inventory

D. Generates proposals

WHY

Software can be configured for it; the discipline is the hard part.

What is the SPRS score?

A. A past performance rating

✓ **B. The self-assessed NIST SP 800-171 cybersecurity score that must be posted before award of a contract involving CUI**

C. A credit rating

D. A size determination

WHY

Honest and low beats absent or inflated; it is a certification.

Why submit a small first invoice early?

A. To earn interest

✓ **B. To find WAWF role, routing, and format problems before a large invoice is due**

C. It is required

D. To trigger an audit

WHY

First invoices take longer than expected for setup reasons.

A startup has a foreign investor. What should it do before proposing?

A. Hide it

✓ **B. Map the ownership and foreign ties, determine which programs they affect, and disclose as required**

C. Withdraw from SBIR

D. Nothing; only classified work cares

WHY

Some ties are disqualifying, some need mitigation, many are fine disclosed. Concealment loses.

Match the registration to its purpose:**CORRECT MATCHES**

SAM.gov	Entity registration, UEI, CAGE, reps and certs
DSIP	DoD SBIR/STTR topics and proposals
WAWF / PIEE	Invoice submission and acceptance
SPRS	NIST 800-171 self-assessment score

WHY

SAM for the entity and reps; DSIP for DoD SBIR proposals; WAWF for invoices; SPRS for the cyber score.

Order the registration sequence:**CORRECT ORDER**

1. Confirm legal name, address, EIN, and bank account match official records
2. Register in SAM; complete validation, CAGE, and reps and certs
3. Register with SBA and DSIP for SBIR/STTR
4. Set up PIEE/WAWF and post the SPRS score
5. Sign up for innovation unit and consortium portals in your domain

WHY

Entity basics first, then SAM validation, then the program-specific registrations, then invoicing and cyber.

What happens if the legal entity holding a contract changes through a merger?

A. Nothing

✓ **B. The government must recognize the successor through novation, which takes months and can be refused**

C. The contract automatically transfers

D. The contract is terminated

WHY

Brief counsel before corporate changes and notify the CO early.

THE STARTUP ON-RAMP: SBIR, OTS, DIU, AND THE VALLEY OF DEATH · LESSON 7

Pricing and Proposing as a Startup: Rates Without a History, Budgets That Survive Review

24 min

How a company with no government cost history builds a defensible price: the minimum indirect rate structure, what goes in fringe, overhead, and G&A when you have never had pools, pricing a fixed-price SBIR versus a cost-type Phase II versus an OT milestone, the cost realism trap of underpricing, the commercialization plan and TABA funds, subcontracting a university on STTR, and how to write a proposal that a government evaluator can score rather than a pitch an investor would fund.

The Startup Pricing Problem

A company that has priced commercially thinks in prices: what the market will pay. Government pricing on R&D and cost-type work thinks in costs: what it will cost you to do the work, built up from labor hours at rates that include your indirect costs, plus a fee. A startup usually has no indirect rates because it has never needed them, no cost history to base them on, and a commercial instinct to price low to win. All three cause trouble. The government evaluates cost realism, so a price that could not possibly cover the work is a risk finding. Rates set too low in the first award become the baseline the government expects in the next. And a company that cannot explain its costs cannot negotiate them. This lesson builds the minimum structure that makes a startup's price defensible.

IN PLAIN TERMS

Think about a plumbing company that pays its plumber \$40 an hour and bills customers \$110. That gap is not greed. It covers the van, the tools, insurance, the office manager, payroll taxes and a modest profit. A company that billed the \$40 wage would go broke. Government cost-based pricing makes that gap explicit: fringe, overhead and general and administrative (G&A) costs are calculated as rates and stacked on salary, so the government can see why the hourly price is what it is.

The Minimum Rate Structure

Three pools are enough for a first award. Fringe: everything you spend on employees beyond salary (payroll taxes, health insurance, retirement match, paid time off), divided by total salaries, gives a fringe rate, typically in the range of a quarter to a third of salary for a small company. Overhead: the cost of supporting technical work (office or lab space, equipment depreciation, software tools, technical management time not charged to a project), divided by direct labor plus its fringe, gives an overhead rate. G&A: everything else that runs the company (executive time not on projects, finance, legal, business

development, insurance), divided by total cost input (all direct and overhead costs), gives a G&A rate. Build each pool from your actual budget for the next year, write down what is in it, and you have provisional rates. The finance module's wrap rate lesson explains how these stack into a billing rate; the point here is that a startup can build them in an afternoon from a spreadsheet and should, before the first cost-type proposal.

From Salary to Loaded Rate

Loaded hourly rate = Base hourly rate × (1 + Fringe) × (1 + Overhead) × (1 + G&A) × (1 + Fee)

An engineer at \$60 per hour base, with fringe of 30 percent, overhead of 40 percent, G&A of 15 percent, and fee of 8 percent, bills at about $\$60 \times 1.30 \times 1.40 \times 1.15 \times 1.08$, roughly \$136 per hour. The multiplier (here about 2.3) is what founders find surprising and what the government expects to see.

Pricing by Instrument

A fixed-price Phase I SBIR: build the cost from hours and rates, add fee, and propose the total; you keep what you do not spend and eat what you overspend, so estimate honestly and include a contingency in the hours, not as a line item. A cost-type Phase II: propose the same build-up, but the government pays actual allowable costs plus the negotiated fee, up to the ceiling; your provisional rates must be defensible because they will be audited and adjusted to actuals. An OT with milestone payments: price each milestone as the cost of reaching it plus a margin, and front-load enough that early milestones fund the ramp; because the FAR does not apply, you can price commercially, but the AO will still ask how you got the number. A subcontract to a prime: the prime will want the same build-up and will apply its own markup on top, so expect your rate to be questioned and be ready to defend each pool.

A LOW R&D PRICE READS AS NOT UNDERSTANDING THE WORK

The instinct to price low to win a first government award is almost always wrong. The evaluator does not reward a low price on R&D; they mark it as unrealistic and question whether you understand the work. The rates you set become your history. And a Phase I priced at half of what the work costs teaches the company that government work loses money, which is the wrong lesson from a program designed to fund you fully. Price the work at what it costs with a fair fee, and compete on the technology and the customer pull.

The Commercialization Plan

At Phase II, the commercialization plan is evaluated alongside the technical approach, and it is where research projects are separated from businesses. It should answer: who are the customers, government and

commercial, and what evidence do you have that they will pay; what is the market and who else serves it; what is the revenue model and the pricing; what funding will carry the company from Phase II to revenue, and from whom; what is the transition path into a program of record (name the office, the mechanism, and the timeline); what intellectual property protects the position; and what the company has already sold. Letters from customers and program offices carry more weight than any projection. A plan that names a PEO and a budget line, cites a user memorandum, and shows commercial revenue is the strongest thing in the proposal.

One sentence to teach your team: **build three indirect pools from your real budget, price the work at what it costs plus a fair fee, write a basis of estimate an evaluator can check, and let the customer letters and the commercialization plan do the winning, because a low price on R&D is read as a risk rather than a strength.**

INTERMEDIATE

The Basis of Estimate: What the Evaluator Reads

Behind every cost number is a basis of estimate: for each task, who works on it (by labor category), for how many hours, and why that many. Evaluators read the BOE to judge whether you understand the work. A BOE that says 2,000 hours of engineering for the whole project is unscorable; one that breaks the project into six tasks with hours per category per task and a sentence explaining each is scoreable and credible. Add materials and equipment with quotes, travel with trips and purposes, subcontracts with the subcontractor's own quote and a note on why them, and other direct costs (cloud compute, test facilities) with a basis. Then apply the rates. A clear BOE also protects you in negotiation: when the CO wants to cut ten percent, you can show which task loses what.

Common Startup Pricing Errors

ERROR	WHAT HAPPENS	FIX
No indirect rates; salary only	Cost realism finding; loses money on award	Build the three pools before proposing
Founder time not charged	Understated cost; founder works free; rates too low for next award	Founders charge direct labor when doing project work
Fee omitted or token	No profit; sets a precedent	Propose a reasonable fee; SBIR allows it
Contingency as a line item	Disallowed or cut	Put realistic risk into the hours estimate
Unallowable costs in pools	Rates rejected on audit; costs disallowed	Code unallowables (entertainment, certain marketing, interest) out of the pools
Subcontractor priced as a lump	Unscoreable; realism question	Include the sub's own quote and build-up

STTR and University Subcontracts

On an STTR, and on many SBIRs, a university does part of the work as a subcontractor. Universities price differently: they have negotiated federal indirect rates (often well above fifty percent on modified total direct costs), they charge for graduate students and faculty summer salary, and their contracting offices move slowly and have firm positions on intellectual property and publication. Get the university's budget from its sponsored programs office, not from the professor, because the professor's estimate will be missing the indirects. Settle the IP agreement early. And remember the STTR work-share minimums: the university's share has to be real and at least thirty percent, while your own has to be at least forty.

TABA and Commercialization Assistance

SBIR agencies may add technical and business assistance funds to awards, at Phase I and larger at Phase II (verify current amounts), for the company to buy outside help with commercialization: market research, intellectual property strategy, regulatory or certification planning, business plan development. It is additional money, requested in the proposal with a named vendor and a plan, and it does not reduce the technical budget. Small companies underuse it. The best uses buy expertise the company lacks and needs for transition: a government market analysis, a data rights strategy, a security certification roadmap, or a fractional government business development advisor.

ADVANCED

Rates as Strategy: What Your First Rates Commit You To

The rates in the first cost-type award become the government's reference for the next one, and the actual rates at year end, once audited, become your history. Senior founders set the first provisional rates from a budget that reflects the company they are building, not the one they have today (the next year's office, the hires planned, the compliance costs coming) and document the basis so the rates hold up. They keep the pools simple until complexity is needed; a single overhead pool serves a company under fifty people. They charge founder and executive time honestly, direct when on projects and G&A when running the company, because the alternative understates cost and shapes a distorted rate history. And they revisit the rates each year against actuals, because a company that grows fast will see its G&A rate fall and its overhead change, and the provisional rates should follow.

Commercial Pricing on OTs and the Line Between Cost and Price

On an OT, nothing requires a cost build-up, and a company with a commercial product can propose a price the way it would to any customer: per unit, per seat, per month, plus a fixed price for the adaptation work. Agreements officers accept commercial pricing when the product is genuinely commercial and the price is supported by what other customers pay. The judgment call is where the commercial product ends and the funded development begins: the adaptation, integration, and test work the government is paying for is usually priced on a cost-plus-margin basis with milestones, while the product itself is priced commercially. Keeping those separate protects the commercial price (the government does not get a cost breakdown of your product) and gives the AO the transparency they need on the development portion. It also sets up the follow-on production negotiation, where the commercial unit price will be the starting point.

The Proposal an Evaluator Can Score

How government proposal evaluation differs from a venture pitch, and what to do about it.

Answer the criteria in order **STRUCTURE**

The solicitation lists evaluation criteria. Use them as your headings, in their order, with their words.

Evaluators score against a sheet. Make the sheet easy to fill in.

Claims need evidence next to them **PROOF**

Every capability claim carries a result, a customer, a test, or a reference. Vision statements score nothing.

A venture deck sells the future. A proposal proves the present.

Risk is discussed, not hidden **CANDOR**

Name the technical risks and the mitigation. Evaluators find the risks anyway; naming them shows understanding.

A proposal with no risks reads as naive.

Compliance is a checklist RULES

Page limits, fonts, required forms, certifications, volume structure. Non-compliance is not evaluated; it is discarded.

Assign one person to the checklist and let no one else touch formatting.

The customer speaks PULL

A user memorandum or a program office letter is the single most persuasive attachment.

Get it before the proposal, not after.

WHAT SENIOR PMS DO DIFFERENTLY

Government evaluators reward proposals that are scoreable, realistic, and evidenced. Founders who win build indirect pools from a real budget before the first cost-type proposal, price at cost plus a fair fee, write a BOE by task and category, keep commercial product pricing separate from funded development, use TABA to buy the expertise they lack, get the customer letter first, and treat the first award's rates as a commitment to be set deliberately.

BUILD ON THIS

- **Wrap Rates & Cost Structure** How the pools stack into a billing rate
- **Unit Economics of a Billable Hour** What the loaded rate has to cover
- **Section L vs. Section M** Answering the criteria in the evaluator's order

Key Terms

Indirect Rates

The percentages applied to direct costs to recover fringe benefits, overhead, and general and administrative expense. A startup with no history uses provisional rates built from a budget.

Fringe

The pool of employee benefit costs (payroll taxes, health insurance, retirement, paid leave) allocated over direct and indirect labor.

Overhead

Costs of supporting the technical work that cannot be charged to one contract: facilities, equipment, technical management, tools. Allocated over direct labor.

G&A

General and administrative: running the company (executive, finance, legal, business development). Allocated over total cost input or another base.

Provisional Billing Rates

Indirect rates approved for interim billing on cost-type work, adjusted to actuals at year end. A startup proposes them from its budget and negotiates them with the CO or DCAA.

Cost Realism

The government's evaluation of whether proposed costs are realistic for the work. An unrealistically low price is a risk finding, not a competitive advantage.

TABA

Technical and Business Assistance: funds SBIR agencies may add to Phase I and II awards for commercialization assistance from outside vendors (market research, IP strategy, regulatory). Verify current amounts.

Commercialization Plan

The Phase II proposal section evaluated for whether the technology will reach paying customers: market, customers, competitors, revenue model, funding, and the transition path.

Fee

Profit on a government cost-type contract, negotiated as a percentage and limited by statute for some contract types. SBIR permits a reasonable fee; OTs negotiate freely.

Basis of Estimate (BOE)

The written rationale for each cost element: who does what for how many hours, why that many, at what rate. What the evaluator reads to decide if the budget is real.

Knowledge Check

Why is a low price on an R&D award a mistake?

A. It is not

✓ **B. The evaluator marks it as unrealistic, the rates become your history, and the company learns the wrong lesson**

C. Low prices are prohibited

D. It triggers an audit

WHY

Compete on technology and customer pull; price the work at what it costs.

What are the three minimum indirect pools?

A. Labor, materials, travel

✓ **B. Fringe, overhead, and G&A**

C. Direct, indirect, fee

D. Sales, marketing, engineering

WHY

Each is built from your budget and divided by its base to give a rate.

What goes in fringe?

A. Office rent

✓ **B. Employee costs beyond salary: payroll taxes, health insurance, retirement, paid leave**

C. Executive salaries

D. Marketing

WHY

Typically a quarter to a third of salary for a small company.

An engineer at \$60 base with fringe 30%, overhead 40%, G&A 15%, and fee 8% bills at roughly:

A. \$65

B. \$90

✓ C. \$136

D. \$200

WHY

$\$60 \times 1.30 \times 1.40 \times 1.15 \times 1.08$ is about \$136; the multiplier is about 2.3.

On a fixed-price Phase I, where should contingency go?

A. As a separate line item

✓ B. In a realistic hours estimate, not as a line item

C. In fee

D. Nowhere

WHY

A contingency line is disallowed or cut; realistic hours are not.

What is the strongest element of a commercialization plan?

A. A large market size number

✓ B. Customer and program office letters, a named transition office and mechanism, and existing sales

C. A long competitor list

D. A five-year projection

WHY

Evidence of pull beats projections.

What is a basis of estimate?

A. A price list

✓ **B. The written rationale for each cost element: who, how many hours, why, at what rate, by task**

C. A government audit

D. The fee justification

WHY

Evaluators read it to judge whether you understand the work; it also protects you in negotiation.

Where should a university subcontract budget come from?

A. The professor's estimate

✓ **B. The university's sponsored programs office, because it includes the negotiated indirect rates the professor's estimate omits**

C. A guess

D. The STTR topic

WHY

University indirects are often well above fifty percent; settle IP early.

What is TABA?

A. A tax credit

✓ **B. Additional SBIR funds for outside commercialization assistance, requested with a named vendor and plan, not deducted from the technical budget**

C. A security clearance

D. A consortium fee

WHY

Underused; best spent on expertise the company lacks for transition.

Match the pool to its base:**CORRECT MATCHES**

Fringe	Total salaries
Overhead	Direct labor plus its fringe
G&A	Total cost input
Fee	Applied last, on total cost

WHY

Fringe over salaries; overhead over direct labor plus fringe; G&A over total cost input.

Order the steps to build a first cost-type price:**CORRECT ORDER**

1. Build fringe, overhead, and G&A pools from the next year's budget
2. Estimate hours by task and labor category
3. Apply the loaded rates to the hours; add materials, travel, subcontracts, ODCs
4. Add a reasonable fee
5. Write the basis of estimate task by task

WHY

Pools from budget, then hours by task, then rates applied, then fee, then the BOE that explains it.

On an OT with a commercial product, how should pricing be structured?

A. A full cost build-up of everything

✓ **B. Commercial pricing for the product, supported by what other customers pay, and cost-plus-margin milestones for the funded adaptation work**

C. Free product, paid support

D. Whatever the AO proposes

WHY

Keeping them separate protects the commercial price and sets up the follow-on production negotiation.

THE STARTUP ON-RAMP: SBIR, OTS, DIU, AND THE VALLEY OF DEATH · LESSON 8

Venture Capital, Dual-Use Strategy, and the Cap Table Rules That Can End Your Eligibility

22 min

How outside capital interacts with defense: why investors have warmed to defense, what a dual-use strategy actually requires, the SBIR ownership rules for venture-backed companies, how foreign investment can disqualify you or force mitigation, affiliation and size, the Office of Strategic Capital and government-backed financing, revenue concentration risk, and how to talk to investors about defense timelines without either overselling or scaring them off.

Why Investors Changed Their Minds

For most of the two decades before the mid-2010s, venture investors avoided defense: long sales cycles, small initial contracts, government data rights, and a system built for large incumbents. That changed. A handful of large outcomes showed the market could produce venture-scale returns; the on-ramps made the early money faster and larger; the geopolitical environment made defense spending durable; and a set of investors built franchises specifically on defense and dual-use technology. Today a startup with a credible defense strategy can raise from specialists and from generalists who follow them. The investors who write those checks understand the timelines, which means they will also test whether you do.

What Dual-Use Really Requires

Dual-use is easy to say and hard to run. The commercial market and the defense market want different things from the same product: commercial customers want speed, features, and price; defense customers want security, integration with legacy systems, sustainment, and documentation. A dual-use company has to keep one product roadmap serving both without letting either customer distort it, run two sales motions with different cycles, meet defense compliance without loading its cost on commercial customers, and manage export controls on a product it sells internationally. The companies that do it well pick the commercial market first (it funds the company and proves the product), enter defense with the product as it is, and resist customizing for defense until a program office is paying for the customization. The companies that struggle let the first enthusiastic defense customer pull the roadmap and end up with a defense-only product and commercial customers who left.

The SBIR Ownership Rules

SBIR requires that the company be more than half owned and controlled by U.S. individuals, by other qualifying small businesses, or, at agencies that have opted in (DoD has), by multiple venture capital, hedge fund, or private equity firms. Under the last route, no single such firm may own more than half, the firms must themselves be majority U.S.-owned and controlled, and the company must disclose the structure in

the proposal. A company controlled by a single fund, or by a foreign-controlled fund, is ineligible. Because eligibility is checked at each award, a financing round between Phase I and Phase II can end a program mid-stream. Founders should run the SBIR eligibility test on every proposed cap table before signing, and should know that convertible instruments and control provisions can count even when the ownership percentages look fine.

IN PLAIN TERMS

Think of an income-qualified apartment. You keep the lease as long as your household meets the limit, and the landlord rechecks it every year. Add a roommate with a large income, and at the next recheck you no longer qualify, even though nothing else changed. SBIR eligibility works the same way: it is checked at every award, so one new investor or one control clause in a term sheet can end eligibility between Phase I and Phase II.

How Capital Structure Affects Eligibility

SITUATION	SBIR	SET-ASIDES AND SIZE	CLEARANCES (FOCI)	PROTOTYPE OT
Founder-majority, U.S. investors as minority	Eligible	Small if affiliation rules are met	No FOCI issue	Non-traditional; eligible
Multiple U.S. VCs together own a majority, none over half	Eligible at DoD with disclosure	Affiliation analysis required; often still small	No FOCI issue	Eligible
One U.S. fund owns a majority	Ineligible	Affiliated with the fund's portfolio; likely large	No FOCI issue	Eligible but may not be non-traditional if affiliates are CAS-covered
Foreign investor with control rights or board seat	Disclosure required; may disqualify depending on country and ties	Affiliation with the foreign parent	FOCI mitigation required for any clearance	Sponsor discretion; often a problem
Foreign-controlled fund holds a majority	Ineligible	Large by affiliation	Clearance unlikely	Usually excluded

EXAMPLE (ILLUSTRATIVE)

A dual-use startup is raising its next round. Today the founders and two U.S. venture firms, Fund A and Fund B, own the company. A new investor, Fund C, offers a term sheet that would give it 52 percent. After that round the founders would hold 22 percent, Fund A 14 percent and Fund B 12 percent ($22 + 14 + 12 + 52 = 100$). A single fund owning more than half makes the company ineligible for SBIR. The founders' counsel restructures the deal so Fund C takes 30 percent: founders 32, Fund A 21, Fund B 17, Fund C 30 (total 100). The three U.S. funds together own $21 + 17 + 30 = 68$ percent and none owns over half, so the company stays eligible at DoD, provided it discloses the structure and the funds are U.S.-controlled.

THE TERM SHEET THAT ENDS YOUR DEFENSE BUSINESS

The term sheet that saves your company can end your defense business. A single large investor taking a majority, a foreign strategic taking a board seat, or a control provision buried in the protective provisions can make you ineligible for SBIR, large by affiliation for set-asides, and a FOCI problem for clearances, all in one signature. Run every term sheet past someone who knows the SBIR eligibility rules, the SBA affiliation rules, and the FOCI rules before you sign, and negotiate the control terms with those rules in view.

Foreign Investment: The Rules Tightened

The 2022 SBIR reauthorization added disclosure requirements for foreign ties: foreign investors, foreign affiliates, personnel with ties to foreign governments, and participation in foreign talent programs. Some relationships disqualify a company outright, particularly ties to countries of concern; others require disclosure and agency judgment. Separately, foreign investment in a company developing critical technologies can require a mandatory CFIUS filing, and a facility clearance requires FOCI mitigation for foreign ownership or influence. None of this prohibits foreign investors, but it makes them a decision with defense consequences. Founders raising from abroad should know which countries, which control rights, and which information access create which problems, and should structure the investment (passive, no board seat, no access to controlled information) to stay clear where possible.

One sentence to teach your team: **every financing decision is also an eligibility decision, because who owns and controls the company determines whether it can hold SBIR awards, compete as small, hold a clearance, or handle controlled information, so the term sheet gets reviewed against those rules before it gets signed.**

INTERMEDIATE

Non-Dilutive Capital as a Financing Strategy

The defense on-ramps are the largest source of non-dilutive capital available to a technology startup. A company that sequences two Phase I awards, a Phase II, and an enhancement with matching funds can bring in several million dollars without selling equity, and a prototype OT can add more. Investors value this: it extends runway, it validates the product with a demanding customer, and it de-risks the defense thesis before they fund it. The discipline is not to let non-dilutive money set the product direction. A company that chases SBIR topics because the money is there ends up with a portfolio of half-built features for different government users. The rule the best companies use: pursue non-dilutive money only where the funded work is on the product roadmap anyway.

How to Talk to Investors About Defense

- **Timelines, honestly** Two to four years from first award to production revenue. Investors who know defense will respect it; those who do not need to hear it before they fund.
- **The transition plan, specifically** Name the program office, the mechanism (Phase III, follow-on production, transition fund), and the budget year. A named PEO is worth more than a market size.
- **Non-dilutive money as validation** What has been awarded, by whom, and what it proves. Not as revenue.
- **Concentration and diversification** Which programs, which services, which commercial customers. Show that no single decision ends the company.
- **Data rights and IP** That the company owns its technology and that the government awards were structured to keep it that way.
- **Eligibility** That the cap table has been checked against SBIR, affiliation, and FOCI rules, and what the round must preserve.

Revenue Concentration and How Acquirers See It

The business module explains how GovCon companies are valued; for a startup, the relevant point is that a company whose revenue is mostly one government program is valued as that program, with a discount for the risk that the program ends. Investors and acquirers look for diversification across programs, services, and customers, and for commercial revenue that proves the product stands on its own. A startup with one large Phase II and nothing else is a research project with a runway; the same startup with two services, a commercial pilot, and a prime subcontract is a business. Diversification also protects the company from the transition failure that ends most defense startups: if one program office says no, the company survives.

The Office of Strategic Capital and Government-Backed Financing

DoD created the Office of Strategic Capital to bring private capital into critical technologies through financial tools (loans, loan guarantees, and partnerships with other federal lending programs) rather than through contracts. It targets areas where the gap is capital intensity (manufacturing, materials, components) rather than R&D, and it works with lenders and investors rather than replacing them. Other federal programs (the Small Business Administration's investment company program, Department of Energy loan programs, and others) intersect with defense-relevant companies. None of these is a substitute for a customer, and a loan against a product with no buyer is a loan. But for a company building physical capacity that a program office wants, government-backed financing can bridge the gap between a production award and the capital to fulfill it. Verify the current programs; they change with each budget.

ADVANCED

Structuring the Round to Preserve Eligibility

Experienced defense-tech counsel negotiates rounds with eligibility in view. They keep any single fund below fifty percent and keep the total U.S.-controlled. They review protective provisions and board composition for control that could trigger affiliation or SBIR control tests, and they document why the investors do not control the company under the SBA rules. They structure foreign participation, where it must exist, as passive economic interest without board seats, information rights over controlled data, or control provisions, and they consider a CFIUS filing where the technology is critical. They record the ownership analysis so the company can answer the disclosure questions accurately and consistently at every award. And they revisit all of it at every round, because the structure that was eligible at seed may not be at Series B.

When Defense Should Not Be the Strategy

Some companies should not pursue defense, and the honest advisor says so. If the product has no plausible defense use without a rebuild, the on-ramps will fund a distraction. If the company's runway is under eighteen months with no commercial revenue, the transition timeline will outlast it. If the founders will not spend time inside the customer, the relationships that transition requires will not form. If the cap table already has foreign control that cannot be unwound, the eligible programs are few. And if the company's culture cannot tolerate compliance, security, and slow customers, defense will make everyone miserable. A company in any of these positions is better served by a prime subcontract for the revenue, or by staying commercial and revisiting defense when the position changes.

Eligibility Review at Each Stage

What to check as the company grows.

Pre-seed and seed **FOUNDERS**

Founder-majority; U.S. investors; no control provisions beyond standard protective ones. Eligible for everything. Set the precedent: every round gets an eligibility review.

Series A **FIRST TEST**

Investors may approach a majority together. Confirm no single fund above half; confirm U.S. control of the funds; review board and protective provisions for affiliation.

Document the analysis for the next SBIR disclosure.

Strategic or foreign investor **CAUTION**

Structure as passive; no board seat; no access to controlled information; consider CFIUS; assess FOCI if a clearance is planned.

Some countries are disqualifying regardless of structure.

Growth round or acquisition **TRANSITION**

Company may become large by size or affiliation; SBIR eligibility may end; plan the graduation and the novation. The small business module's graduation lesson applies.

WHAT SENIOR PMS DO DIFFERENTLY

Government program managers who work with venture-backed companies ask about the cap table early, because they have seen awards collapse on eligibility. Founders who build durable defense businesses review every term sheet against SBIR, affiliation, and FOCI rules, keep commercial revenue as the base, pursue non-dilutive money only on the roadmap, diversify across programs and services, tell investors the real timeline and the named transition path, and know when defense is the wrong strategy for the company they are.

BUILD ON THIS

- **Affiliation** How investor control makes you large
- **How GovCon Companies Are Valued** What concentration does to value
- **Facility and Personnel Security** FOCI and clearances

Key Terms

Dual-Use

A product or company serving both commercial and defense markets. The model the on-ramps favor and most defense-focused investors require.

Majority VC-Owned Eligibility

The SBIR rule allowing companies majority-owned by multiple venture capital, hedge fund, or private equity firms to compete at agencies that opt in, provided no single such firm owns more than half and the firms are U.S.-controlled. DoD has opted in.

Foreign Ownership Disclosure

The requirement, strengthened by the 2022 SBIR reauthorization, to disclose foreign investment, ties, and affiliations, with disqualification for certain relationships. Verify current rules.

CFIUS

The Committee on Foreign Investment in the United States: reviews foreign investments in U.S. businesses for national security risk, with mandatory filings for certain investments in companies working on critical technologies.

FOCI Mitigation

Arrangements (board resolutions, security control agreements, proxy agreements, voting trusts) that allow a company with foreign ownership or influence to hold a facility clearance.

Affiliation

The SBA rule under which entities that control or are controlled by others, or share common control, are counted together for size. Investor control rights are the usual trigger for startups.

Office of Strategic Capital (OSC)

The DoD office created to attract and scale private capital into critical technologies, using loans, loan guarantees, and partnerships rather than contracts. Verify current programs.

Revenue Concentration

The share of revenue from one customer or program. Investors and acquirers discount companies where one government program is most of the revenue.

Non-Dilutive Capital

Money that does not sell equity: SBIR awards, prototype OTs, grants, prizes. Defense on-ramps are the largest source of it for technology startups.

Runway

Months of operation the company can fund at its burn rate. The number that has to exceed the transition timeline.

Knowledge Check

Why did venture investors warm to defense?

- A. Government mandated it
- ✓ **B. Large outcomes proved venture-scale returns, on-ramps made early money faster, spending became durable, and specialist investors built franchises**
- C. Defense contracts became risk-free
- D. Commercial markets disappeared

WHY

The investors who write defense checks understand the timelines and will test whether you do.

What is the most common way dual-use companies fail at it?

- A. Too much commercial revenue
- ✓ **B. Letting the first enthusiastic defense customer pull the roadmap until the product is defense-only and commercial customers leave**
- C. Refusing defense work
- D. Over-investing in security

WHY

Enter defense with the product as it is; customize only when a program office pays.

Under the majority VC-owned SBIR route at DoD, which is required?

- A. A single fund owning a majority
- ✓ **B. No single VC, hedge fund, or PE firm owns more than half; the firms are U.S.-controlled; the structure is disclosed**
- C. Foreign funds are permitted if passive
- D. The founders own nothing

WHY

A company controlled by one fund or a foreign-controlled fund is ineligible.

Why can a financing between Phase I and Phase II end an SBIR program?

A. SBIR prohibits raising money

✓ **B. Eligibility is checked at each award, and the new cap table may fail the ownership or control tests**

C. Investors must approve Phase II

D. Phase II requires a bank loan

WHY

Run the eligibility test on every proposed cap table before signing.

What did the 2022 SBIR reauthorization add?

A. Larger awards only

✓ **B. Disclosure requirements for foreign investors, affiliates, personnel ties, and talent programs, with disqualification for certain relationships**

C. A ban on venture investors

D. Mandatory Phase III

WHY

Some ties disqualify outright; others require disclosure and judgment.

How should a necessary foreign investment be structured to minimize defense consequences?

A. With a board seat and information rights

✓ **B. As a passive economic interest without board seats, control provisions, or access to controlled information, with CFIUS considered**

C. Through a U.S. shell

D. It cannot be

WHY

Structure helps for many countries; some are disqualifying regardless.

What is the rule for pursuing non-dilutive money?

A. Take every award available

✓ B. Pursue it only where the funded work is on the product roadmap anyway

C. Only SBIR, never OTs

D. Only after Series A

WHY

Chasing topics produces half-built features for different users.

How do acquirers view a startup whose revenue is mostly one government program?

A. As highly valuable

✓ B. As that program, discounted for the risk it ends

C. As a commercial company

D. As ineligible

WHY

Diversification across programs, services, and commercial customers is what makes it a business.

What does the Office of Strategic Capital do?

A. Awards SBIR contracts

✓ B. Brings private capital into critical technologies through loans, guarantees, and partnerships rather than contracts

C. Runs CSOs

D. Certifies investors

WHY

Targets capital-intensive gaps; not a substitute for a customer.

Match the rule set to what it governs:**CORRECT MATCHES**

SBIR ownership rules	Eligibility for SBIR/STTR awards
SBA affiliation	Whether the company counts as small
FOCI	Whether the company can hold a facility clearance
CFIUS	National security review of foreign investment

WHY

SBIR ownership rules govern award eligibility; affiliation governs size; FOCI governs clearances; CFIUS governs foreign investment review.

Order the eligibility review as a company grows:**CORRECT ORDER**

1. Seed: founder-majority, U.S. investors, standard provisions
2. Series A: check no single fund above half, U.S. control, and board provisions for affiliation
3. Strategic or foreign investor: structure passively; consider CFIUS and FOCI
4. Growth or acquisition: plan for becoming large and for novation

WHY

Founder-majority at seed, the first real test at Series A, caution with strategic or foreign investors, and graduation planning at growth.

When should a company not pursue defense?

A. Never; every company should

✓ **B. When the product has no plausible defense use without a rebuild, runway is too short for the transition timeline, founders will not engage the customer, or foreign control cannot be unwound**

C. When it has commercial revenue

D. When it is venture-backed

WHY

A prime subcontract for revenue or staying commercial may serve better until the position changes.

**THE STARTUP ON-RAMP: SBIR, OTS, DIU, AND THE VALLEY OF DEATH · MODULE
ASSESSMENT**

Module Assessment

Pass this to unlock the next skill level in the app.

Knowledge Check

The valley of death exists because:

- A. Prototypes usually fail
- ✓ B. R&D money that funds prototypes and procurement money that buys at scale are controlled by different organizations on different budget timelines

- C. Innovation units are underfunded
- D. Startups cannot get clearances

WHY

The finder's money ends when the prototype succeeds; the buyer's budget was set two years earlier.

A non-traditional defense contractor is defined by:

- A. Being under five years old
- ✓ B. Not having performed on a CAS-covered DoD contract for at least a year

- C. Having fewer than 50 employees
- D. Being venture-backed

WHY

The definition unlocks prototype OT eligibility and other flexibilities.

SBIR Phase III allows:

- A. A larger Phase II
- ✓ B. Any agency to award the SBIR firm sole source, in any amount, with non-SBIR money, for work deriving from the SBIR effort

- C. Only the funding agency to award follow-on work
- D. A grant with no deliverables

WHY

SBIR itself is treated as the competition. Most awardees never use it.

During the SBIR data rights protection period, the government may:

A. Release data to competitors

✓ B. Use the data for government purposes but not disclose it outside government or use it for manufacture by others

C. Do anything

D. Nothing

WHY

Stronger than most FAR defaults, provided deliverables are marked correctly.

Follow-on production without competition after a prototype OT requires:

A. Only successful completion

✓ B. A competitively awarded prototype, successful completion, and prior notice in the solicitation and agreement

C. A J&A

D. A consortium membership

WHY

All three conditions; the language is the term most often missing.

Data rights in an OT are:

A. Governed by DFARS 7013 automatically

✓ B. Whatever the agreement says; there is no default

C. Always unlimited to the government

D. Always retained by the company

WHY

Negotiate from the commercial position; separate deliverables from background technology.

Before investing in a CSO solution brief, a company should determine:

A. The evaluators' names

✓ B. Which program office is the transition partner and what money it has committed

C. The page count

D. The consortium fee

WHY

Problem statements with a committed partner transition; the rest produce demos.

The buyer for a transition is defined by:

A. A champion who loves the product

✓ B. A program office with the mission, a budget line that can pay, and a requirement the product satisfies or can be written to

C. The innovation unit

D. The end user alone

WHY

Champions rotate; budget lines survive them.

Which requirement blocks a cost-type award but not a fixed-price Phase I?

A. SAM registration

✓ B. An adequate accounting system reviewed by DCAA or DCMA

C. A CAGE code

D. WAWF registration

WHY

Start the accounting configuration before the Phase II proposal.

A low price on an R&D proposal is typically read by evaluators as:

A. A competitive strength

✓ **B. A cost realism risk suggesting the company does not understand the work**

C. Neutral

D. A reason for immediate award

WHY

Price at cost plus a fair fee; compete on technology and customer pull.

Under DoD's SBIR rules for venture-backed companies:

A. One fund may own a majority

✓ **B. Multiple U.S.-controlled VC, hedge, or PE firms may together own a majority if no single one owns more than half and the structure is disclosed**

C. Foreign funds may control if passive

D. VC ownership is prohibited

WHY

Eligibility is checked at each award, so a financing can change it.

The rule for pursuing non-dilutive defense money is:

A. Take every available award

✓ **B. Pursue it only where the funded work is on the product roadmap anyway**

C. Only SBIR, never OTs

D. Only after commercial revenue exceeds \$10M

WHY

Chasing topics for the money produces a portfolio of half-built features.